

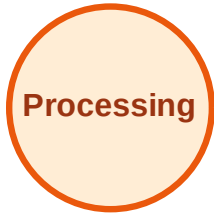
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



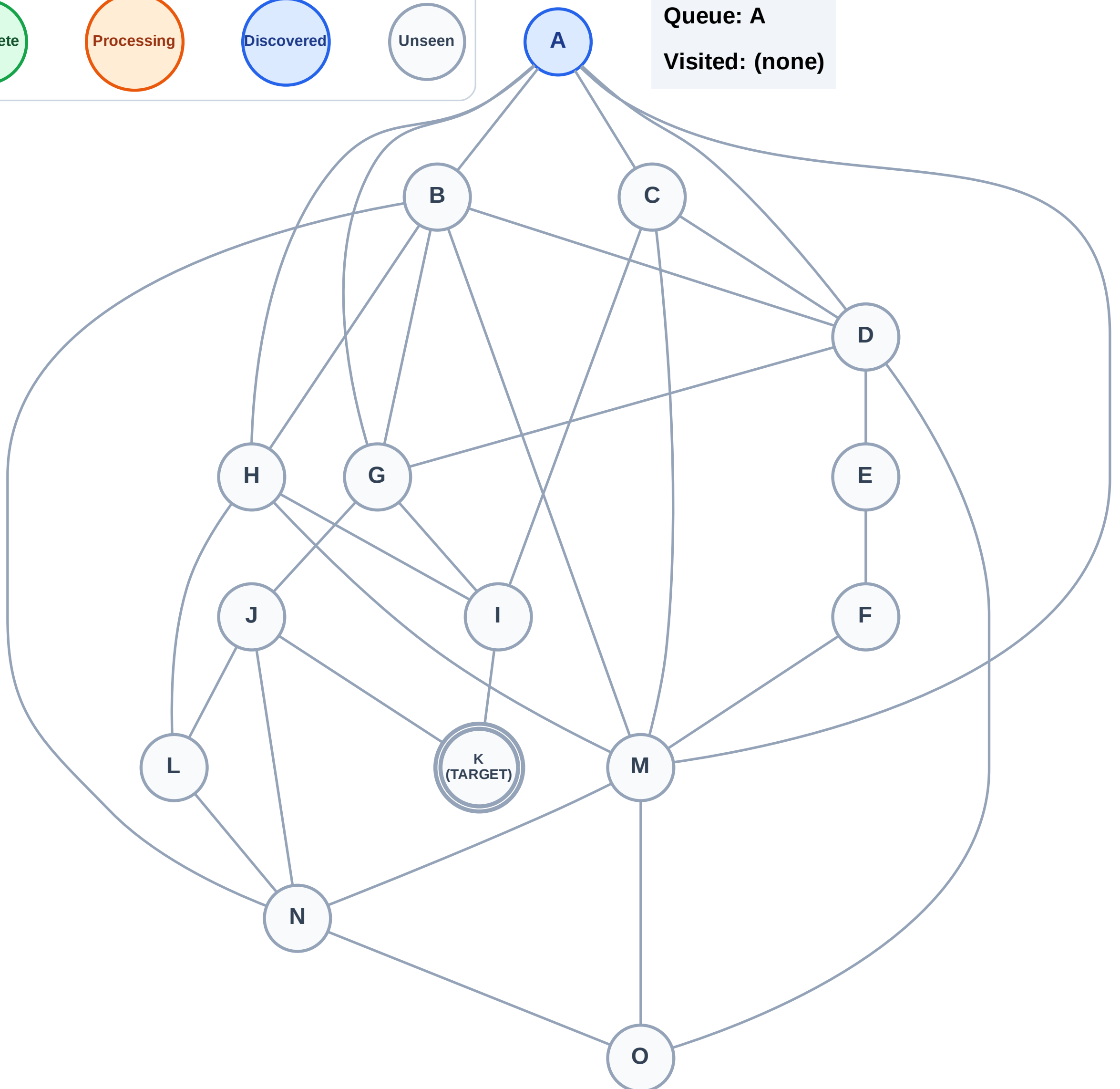
Discovered



Unseen

Queue: A

Visited: (none)



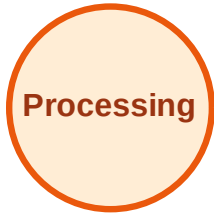
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



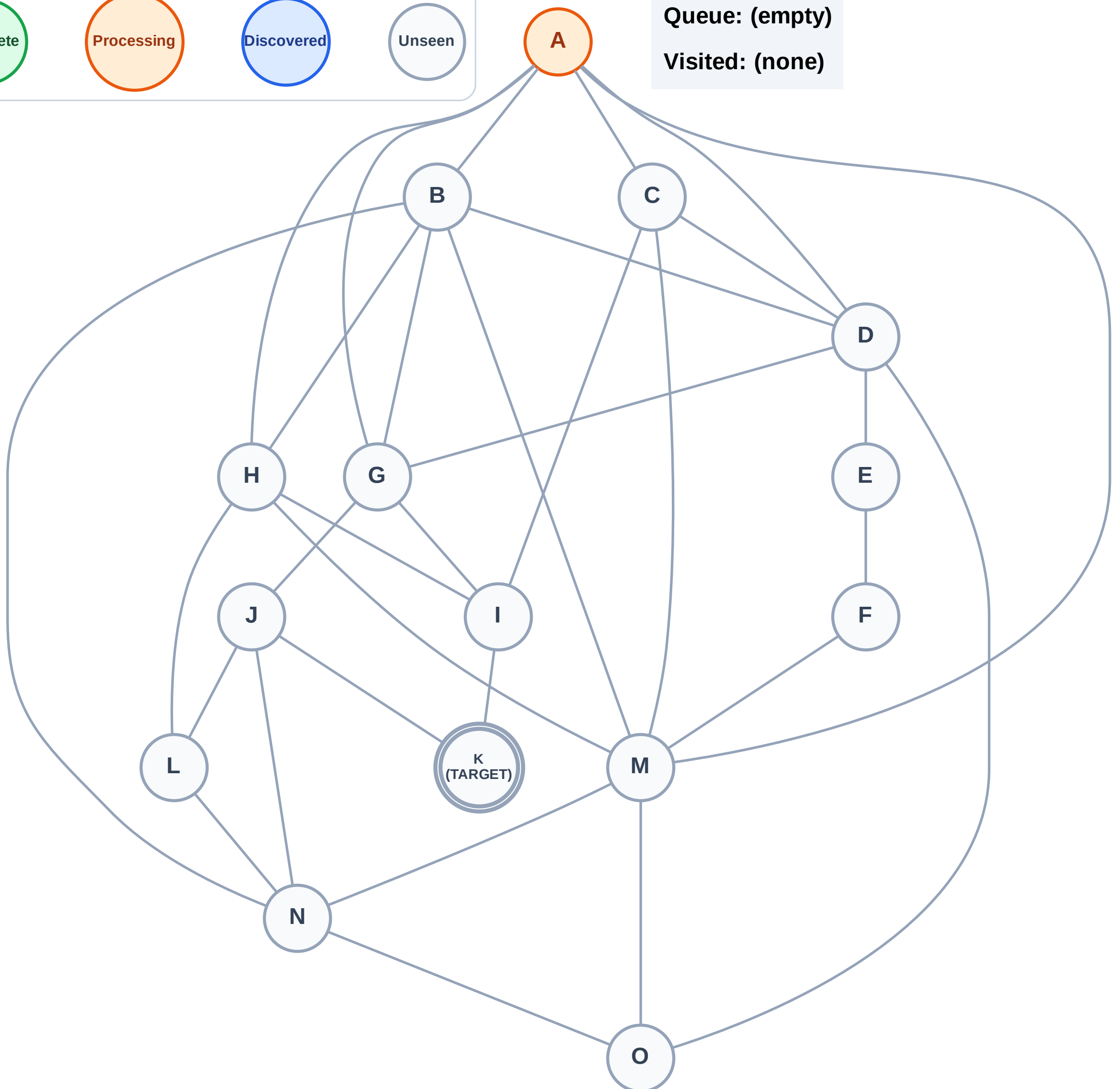
Discovered



Unseen

Queue: (empty)

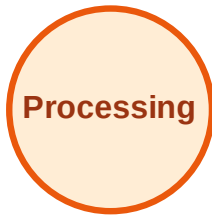
Visited: (none)



BFS Traversal

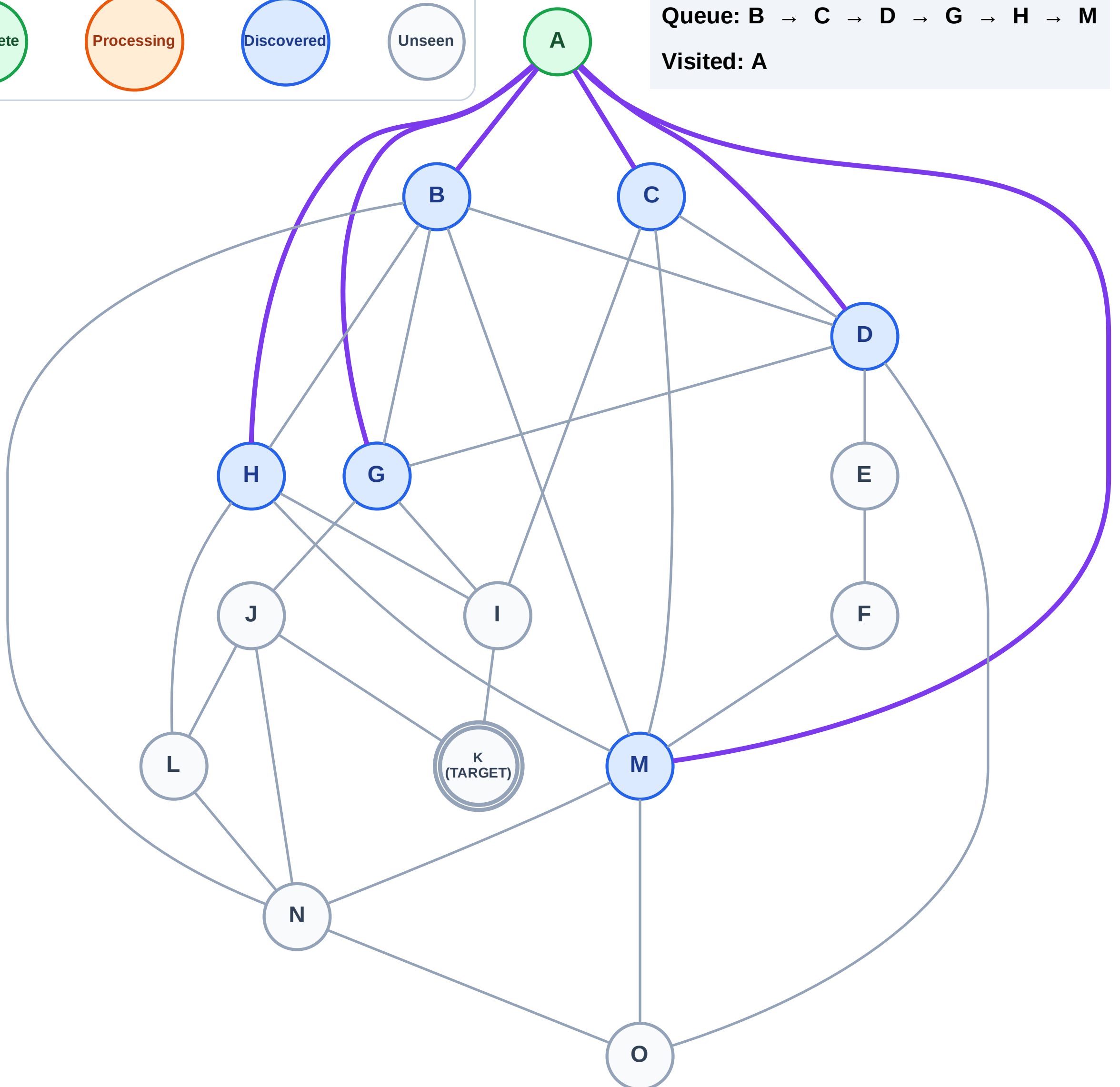
Step 3: Discover B, C, D, G, H, M from A

Legend



Queue: B → C → D → G → H → M

Visited: A



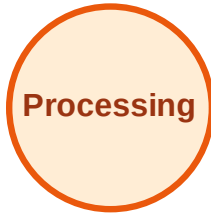
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



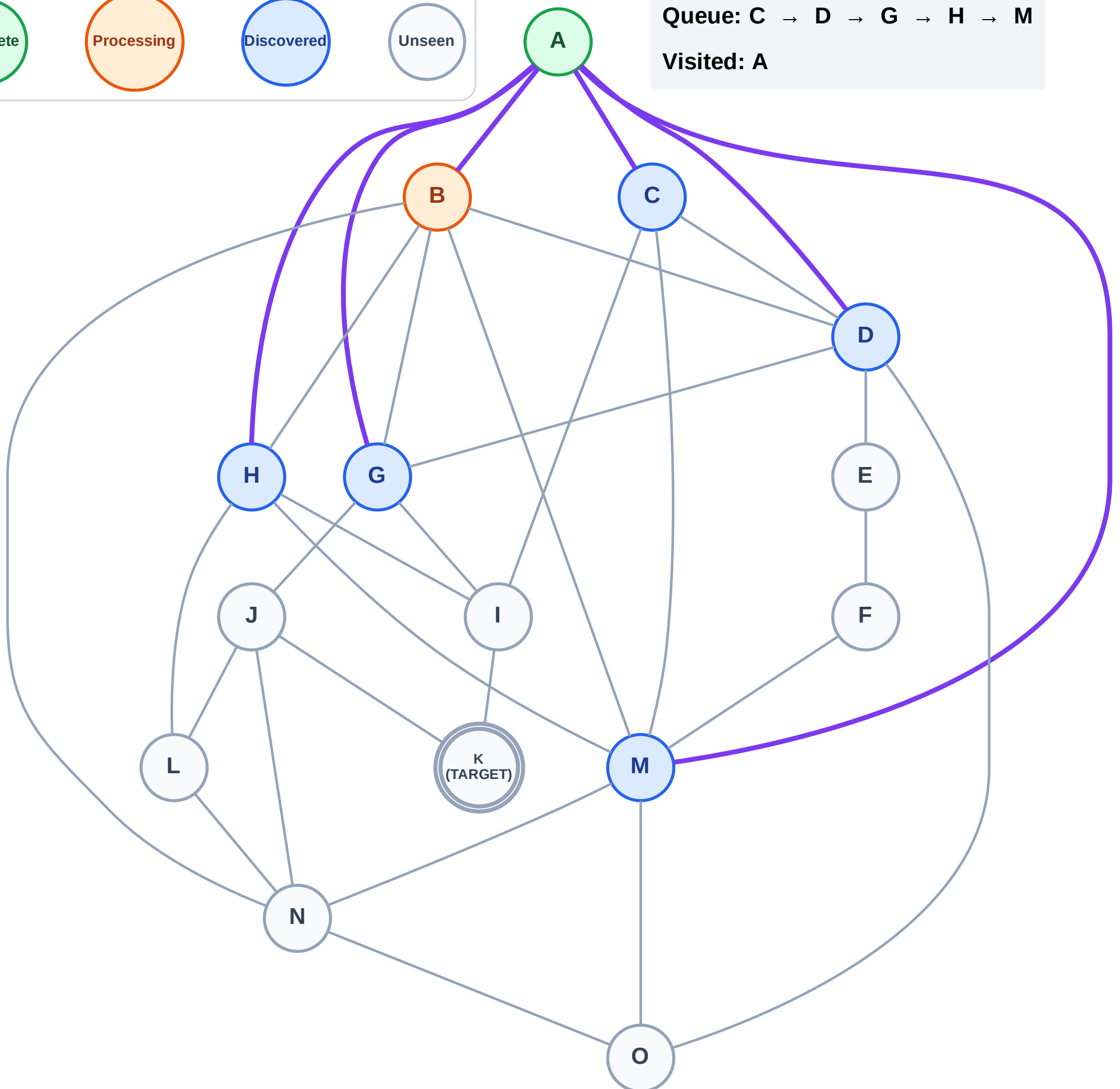
Discovered



Unseen

Queue: C → D → G → H → M

Visited: A



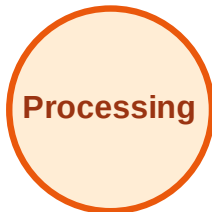
BFS Traversal

Step 5: Discover N from B

Legend



Complete



Processing



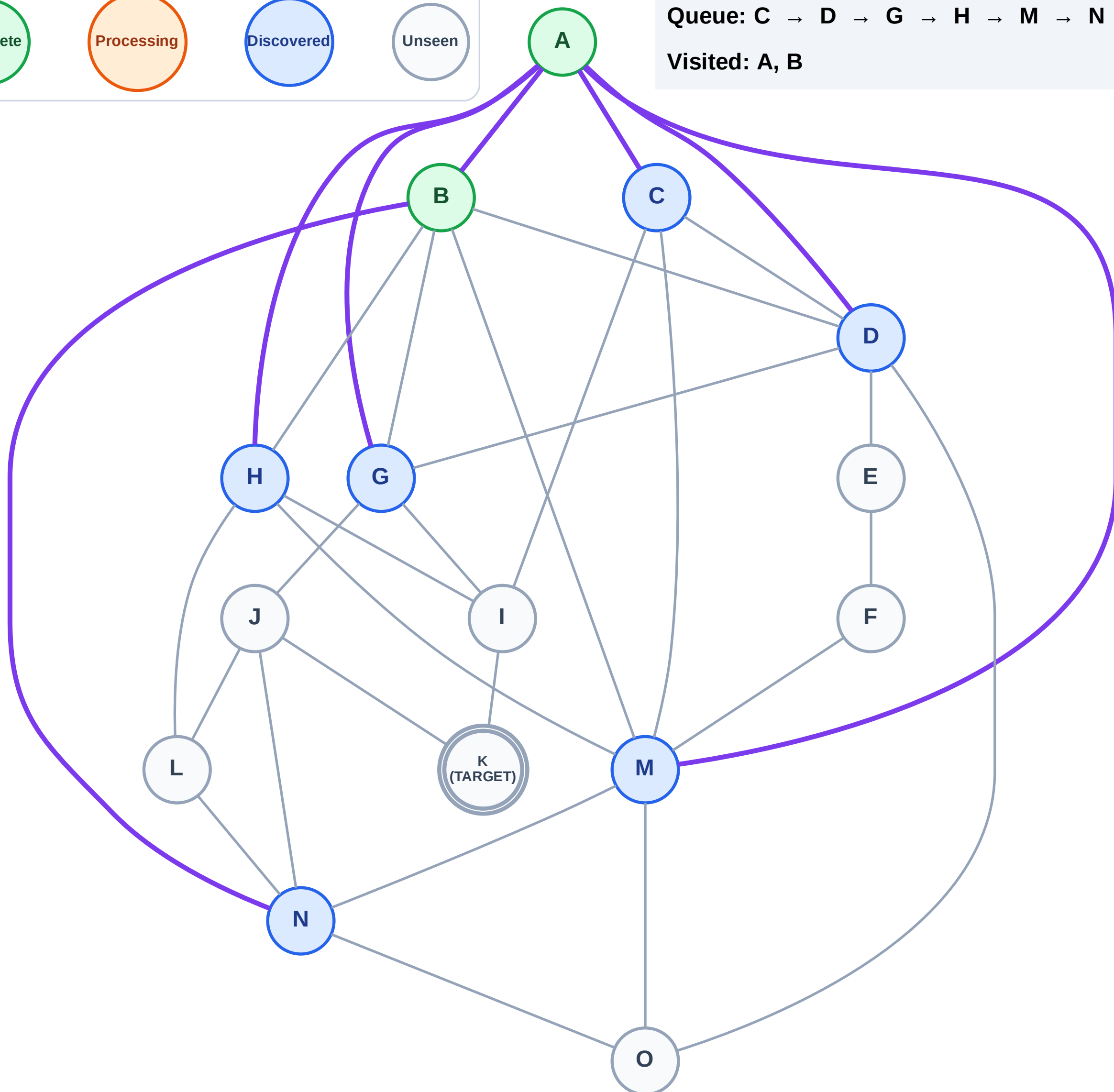
Discovered



Unseen

Queue: C → D → G → H → M → N

Visited: A, B



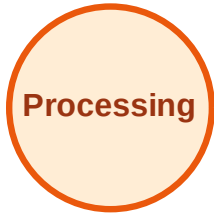
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



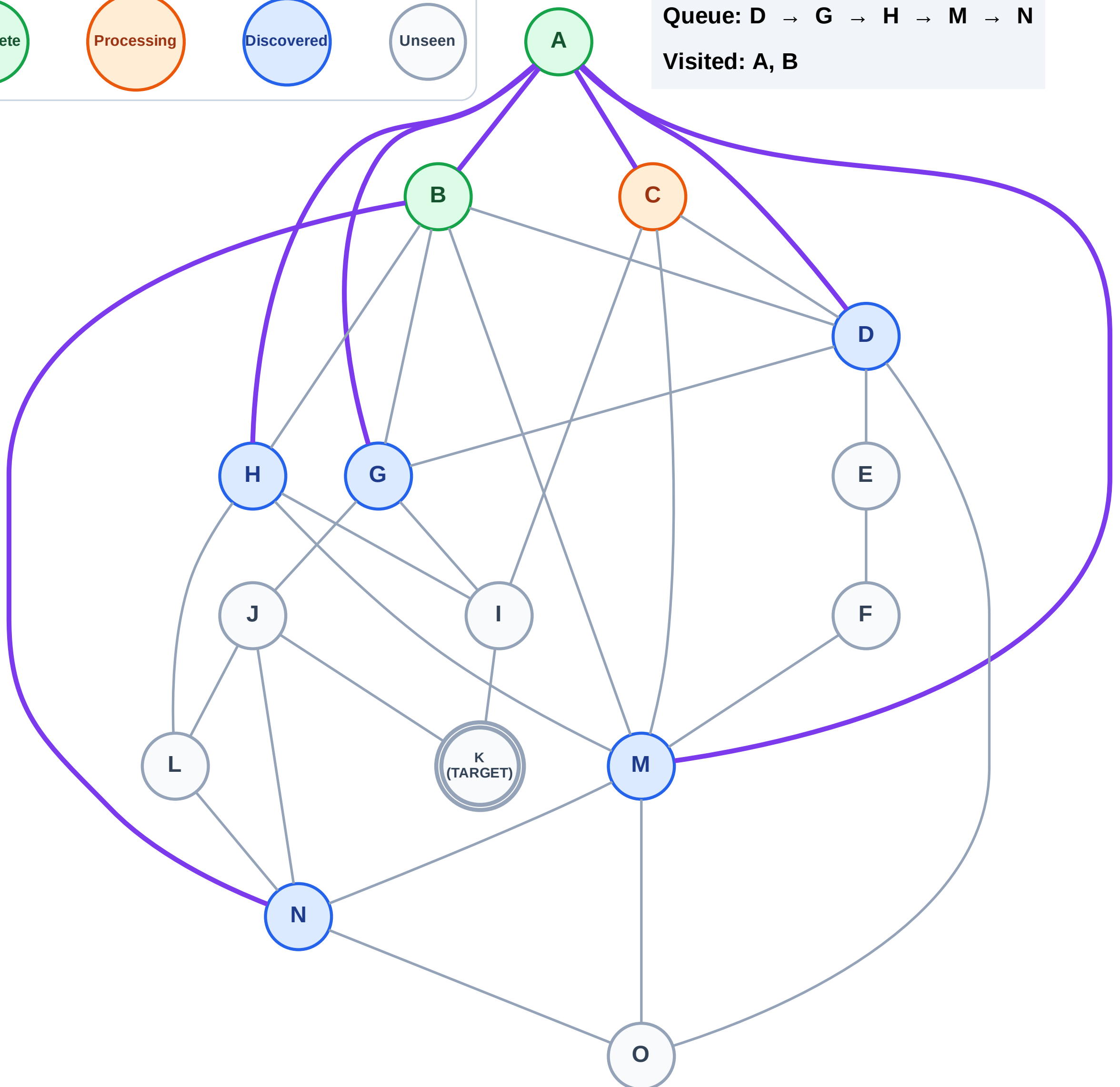
Discovered



Unseen

Queue: D → G → H → M → N

Visited: A, B



BFS Traversal

Step 7: Discover I from C

Legend



Complete



Processing



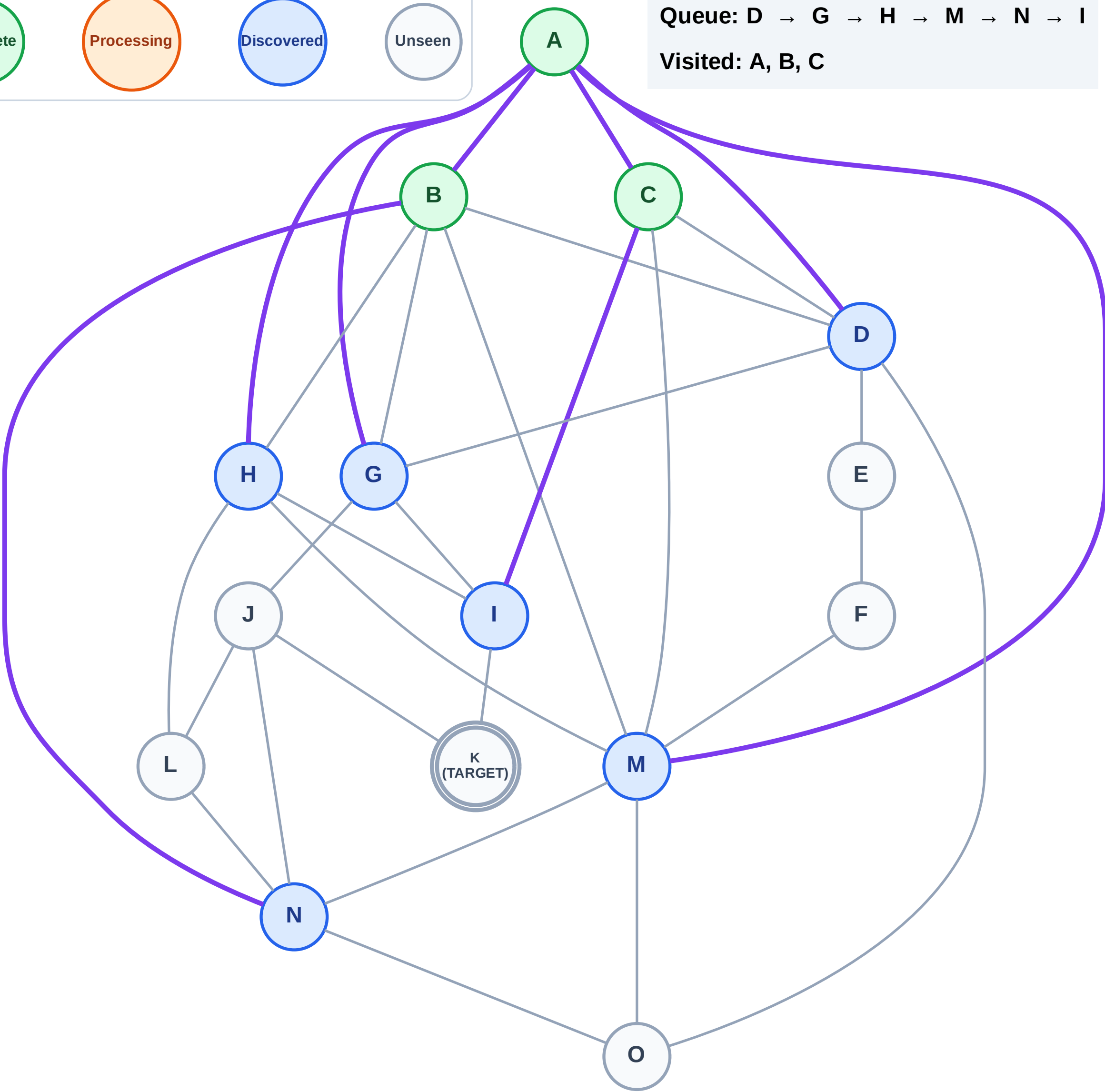
Discovered



Unseen

Queue: D → G → H → M → N → I

Visited: A, B, C



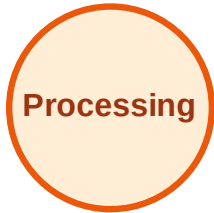
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



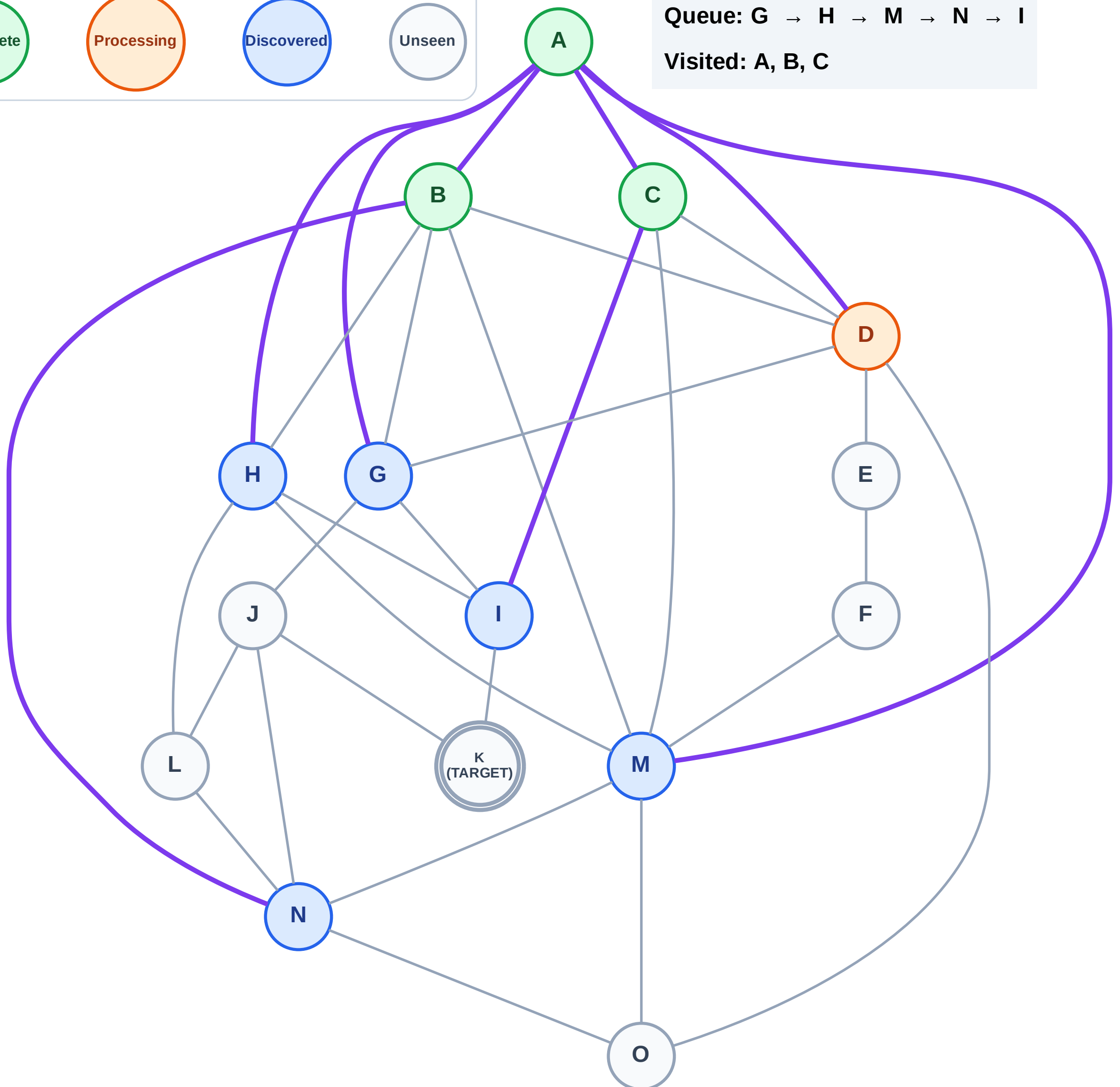
Discovered



Unseen

Queue: G → H → M → N → I

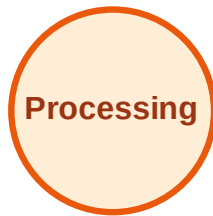
Visited: A, B, C



BFS Traversal

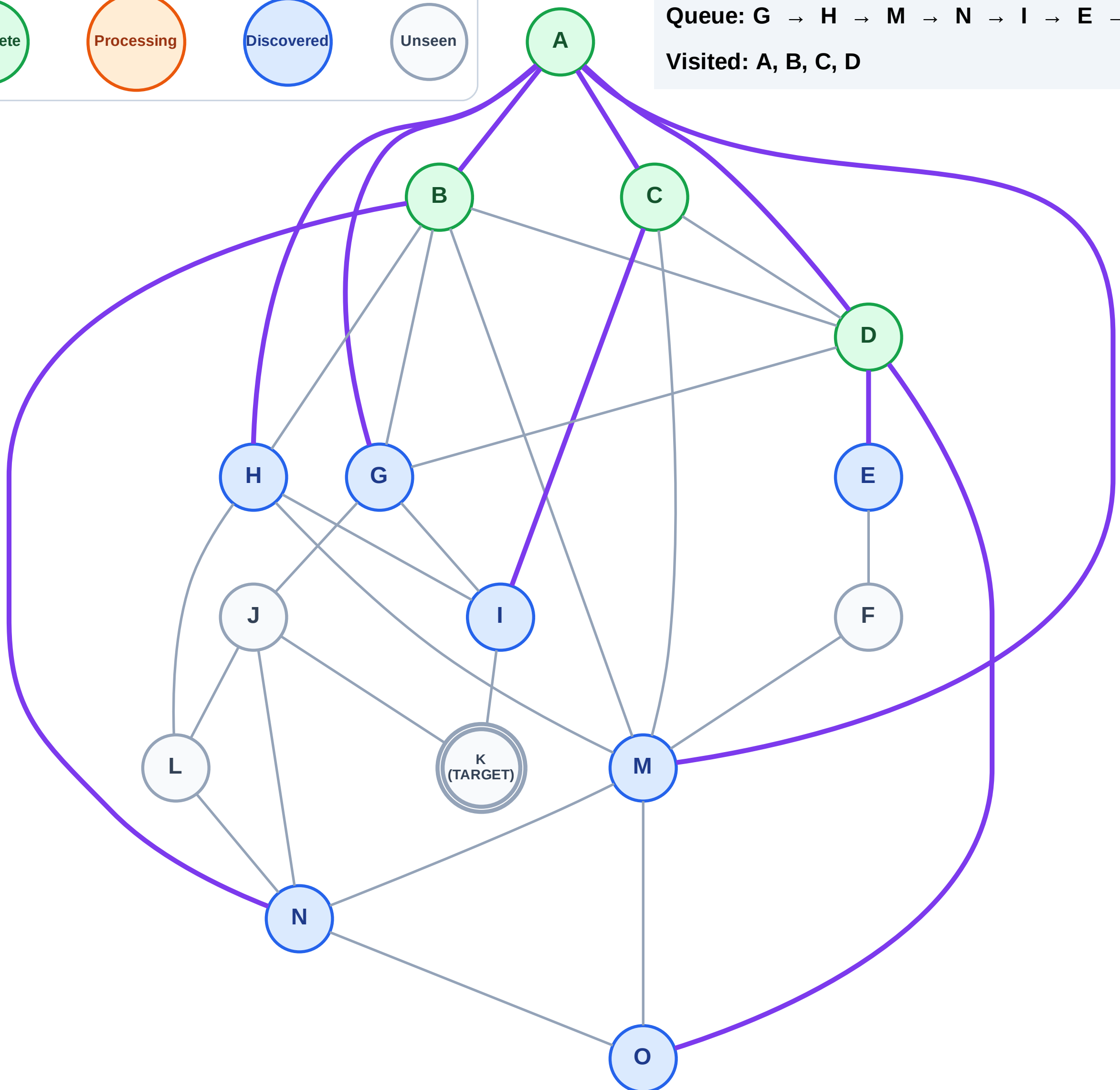
Step 9: Discover E, O from D

Legend



Queue: G → H → M → N → I → E → O

Visited: A, B, C, D



BFS Traversal

Step 10: Process node G

Legend

Complete

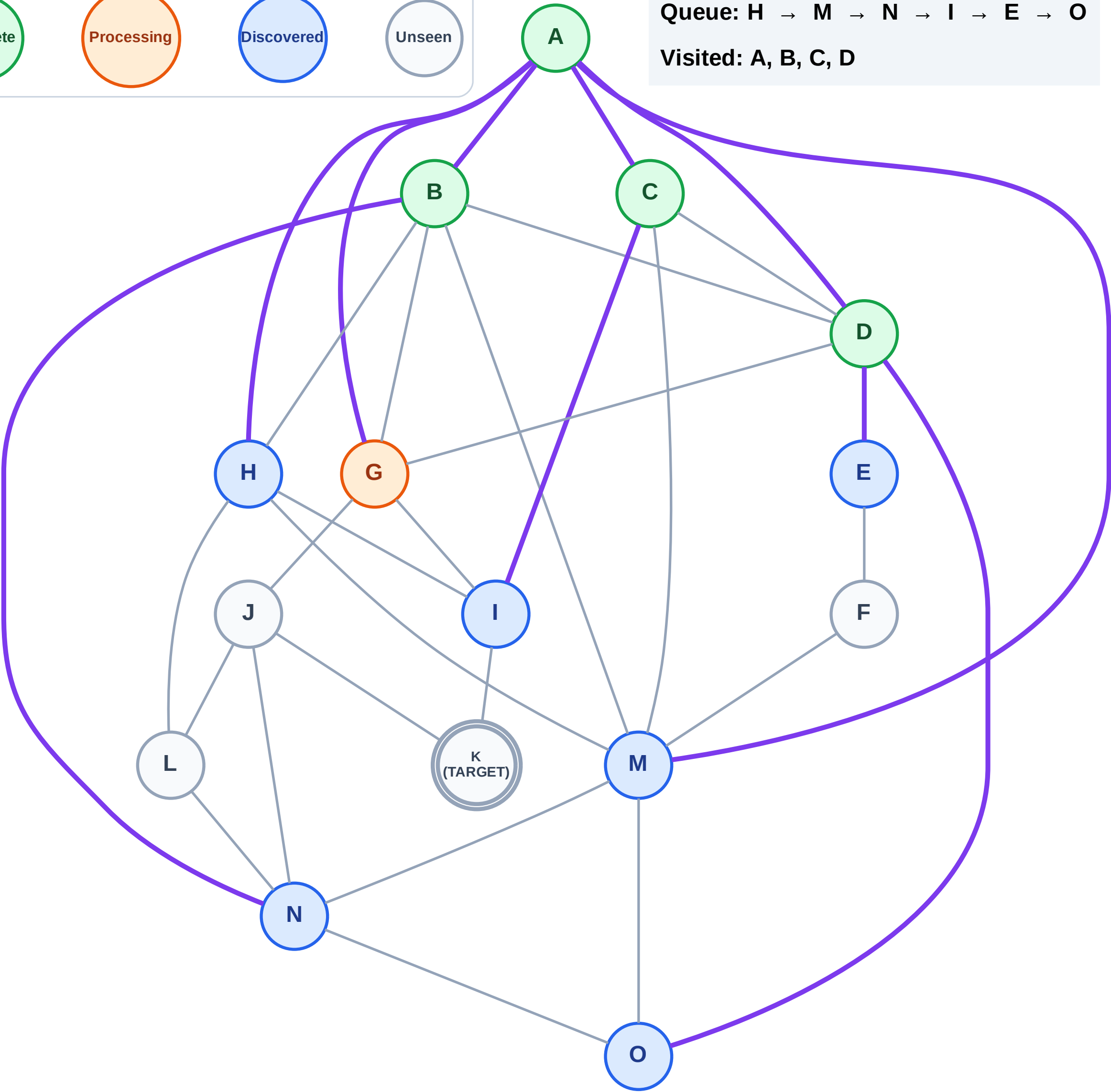
Processing

Discovered

Unseen

Queue: H → M → N → I → E → O

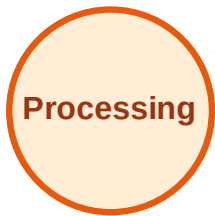
Visited: A, B, C, D



BFS Traversal

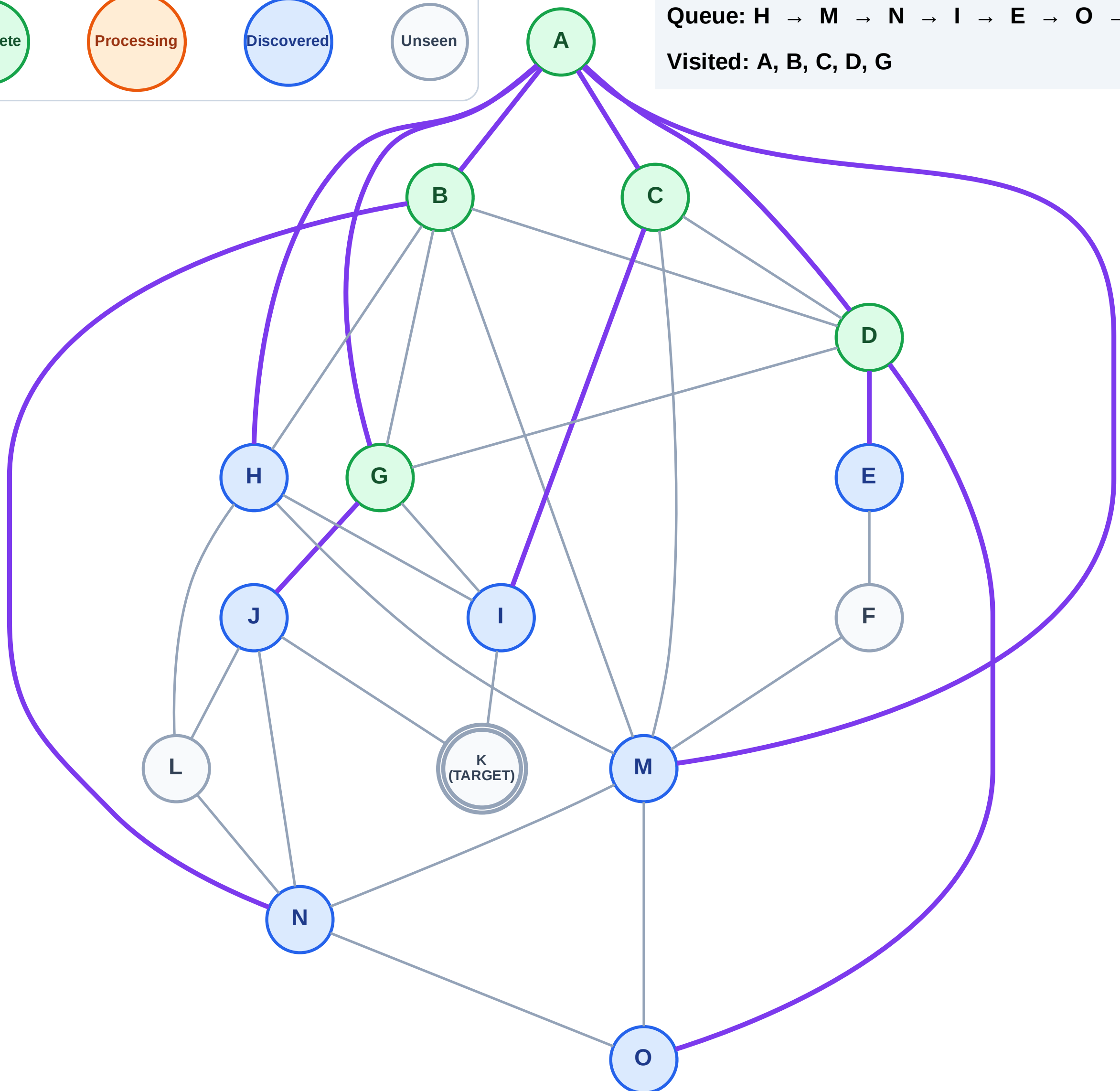
Step 11: Discover J from G

Legend



Queue: H → M → N → I → E → O → J

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node H

Legend

Complete

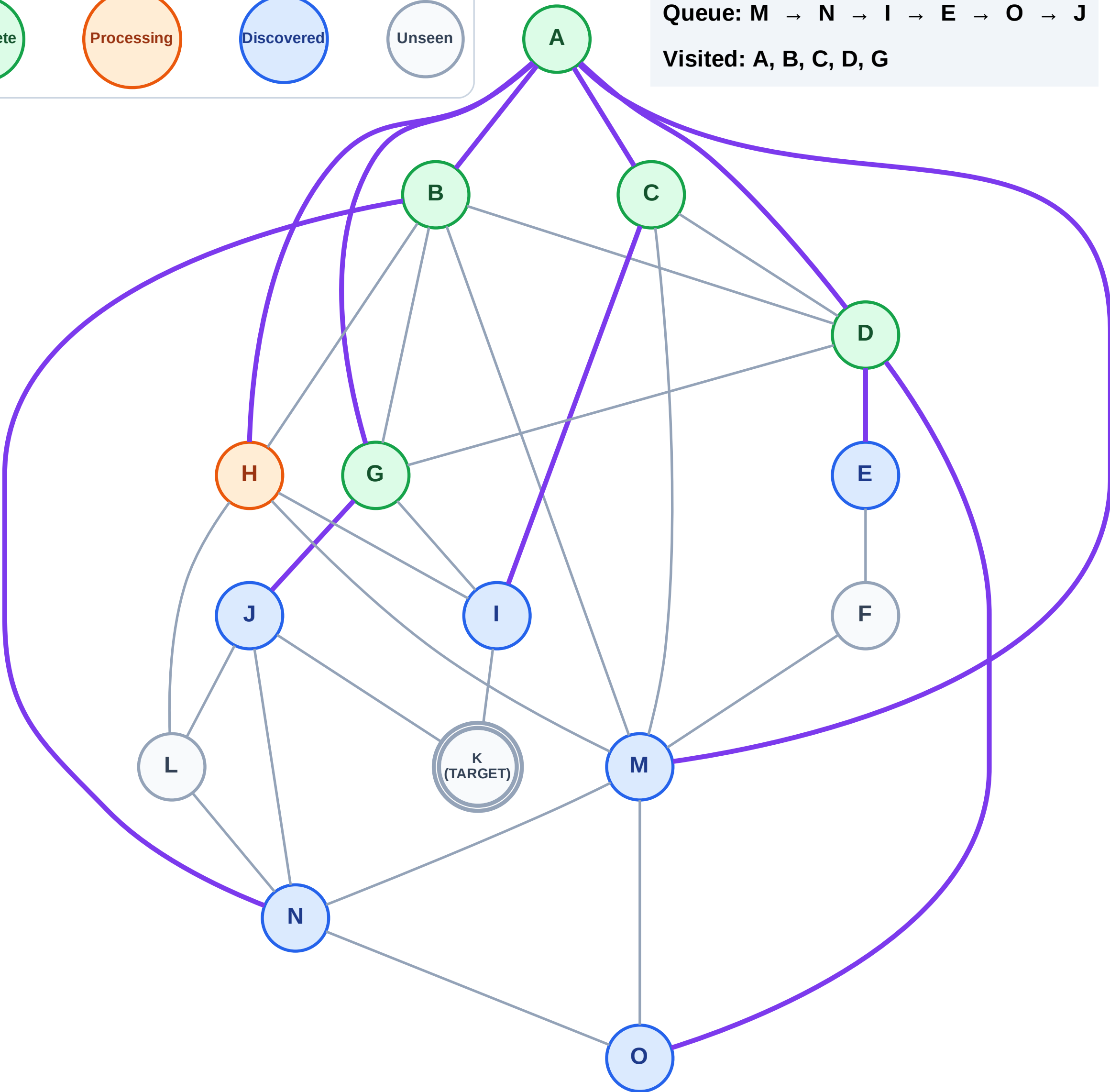
Processing

Discovered

Unseen

Queue: M → N → I → E → O → J

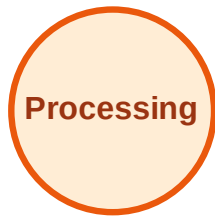
Visited: A, B, C, D, G



BFS Traversal

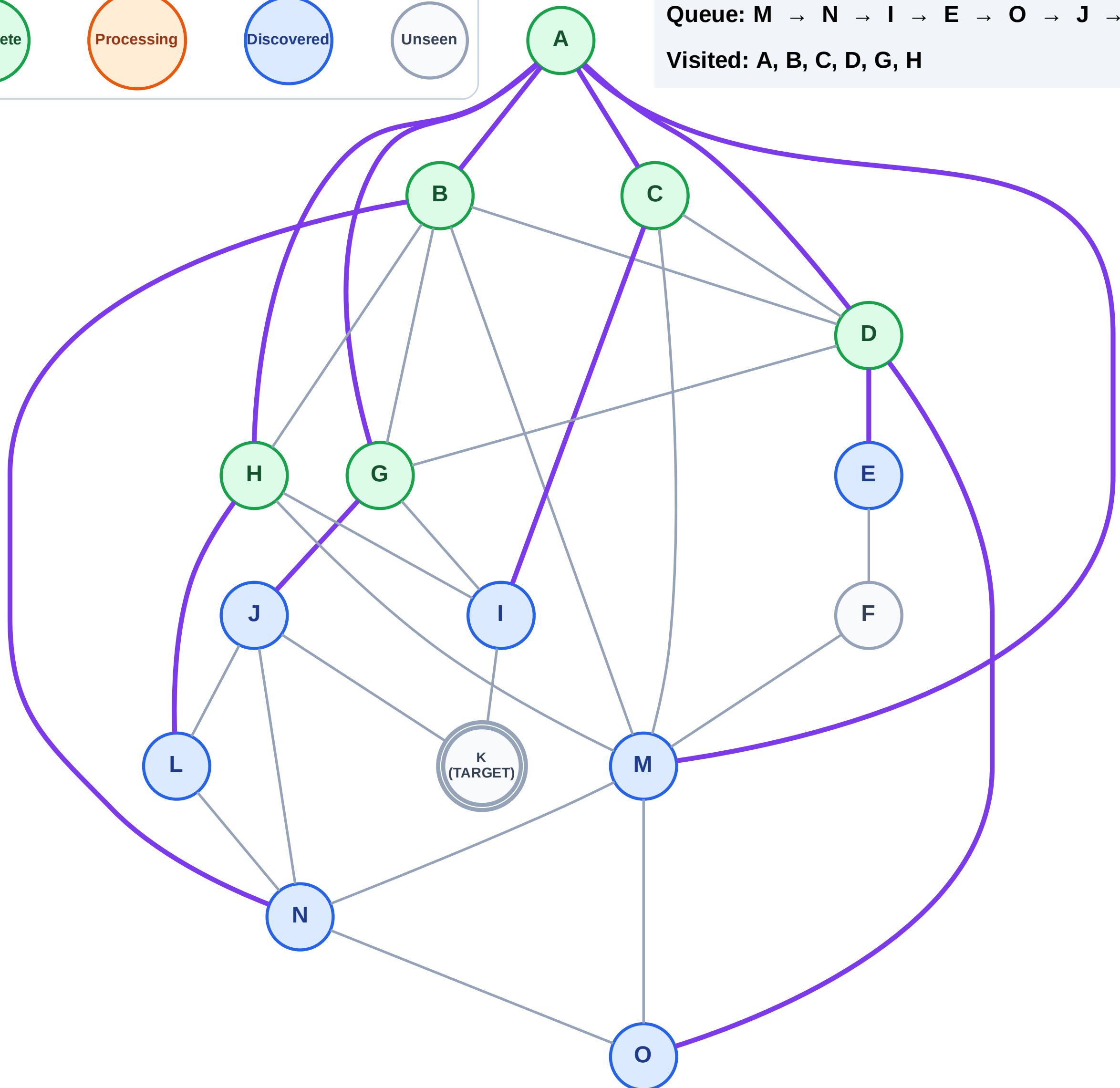
Step 13: Discover L from H

Legend



Queue: M → N → I → E → O → J → L

Visited: A, B, C, D, G, H



BFS Traversal

Step 14: Process node M

Legend

Complete

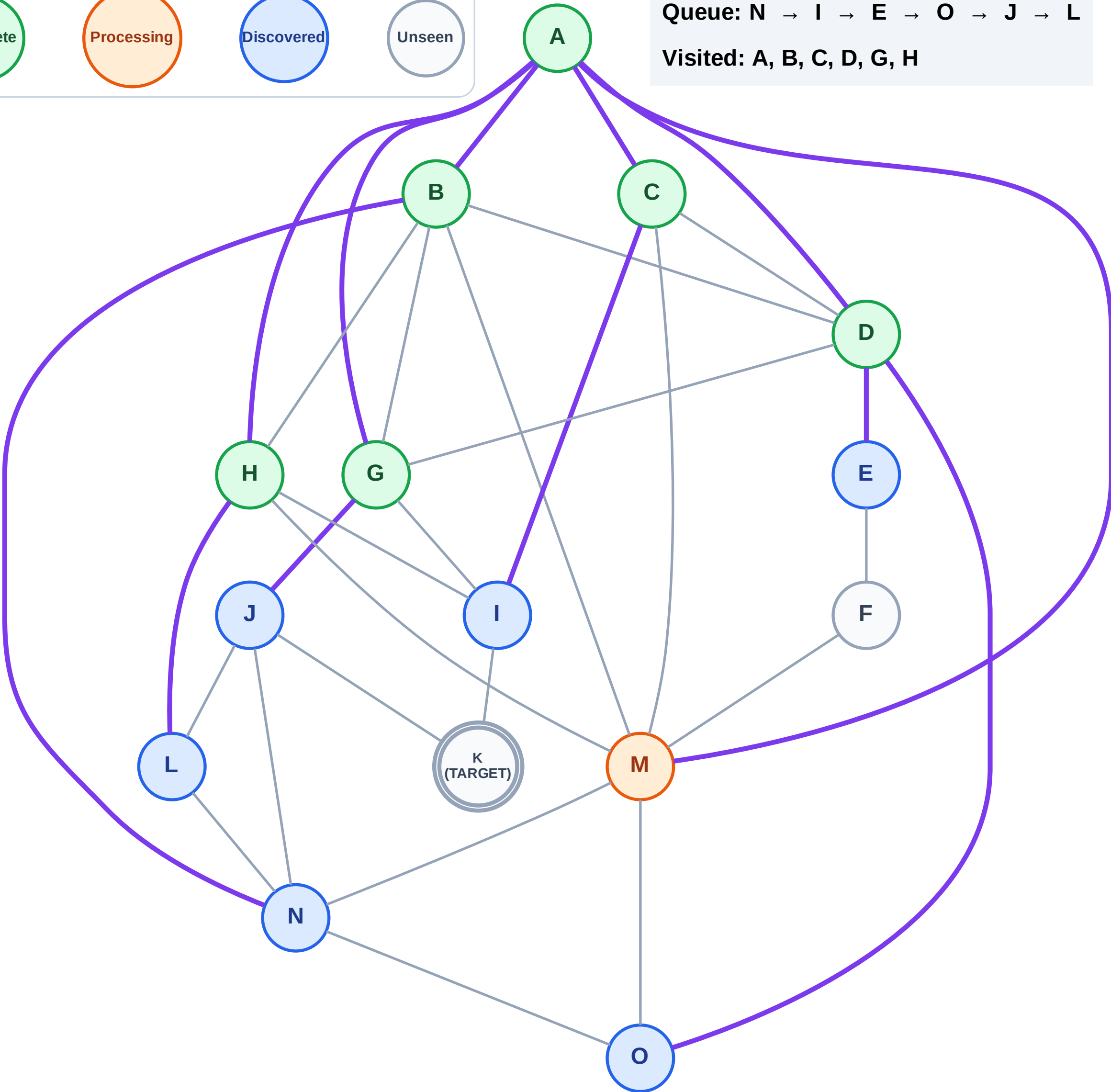
Processing

Discovered

Unseen

Queue: N → I → E → O → J → L

Visited: A, B, C, D, G, H



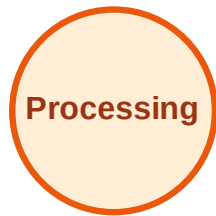
BFS Traversal

Step 15: Discover F from M

Legend



Complete



Processing



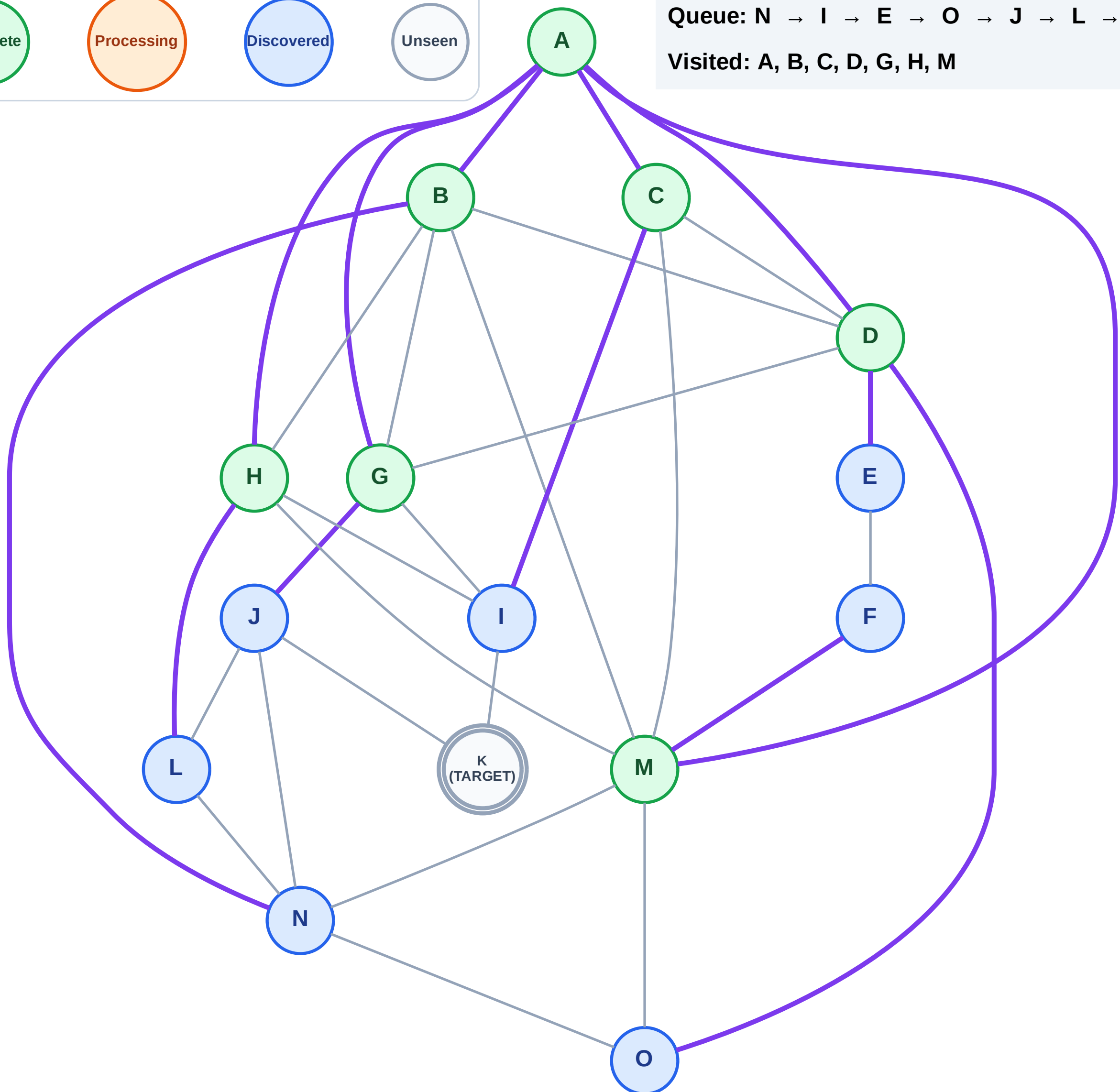
Discovered



Unseen

Queue: N → I → E → O → J → L → F

Visited: A, B, C, D, G, H, M



BFS Traversal

Step 16: Process node N

Legend

Complete

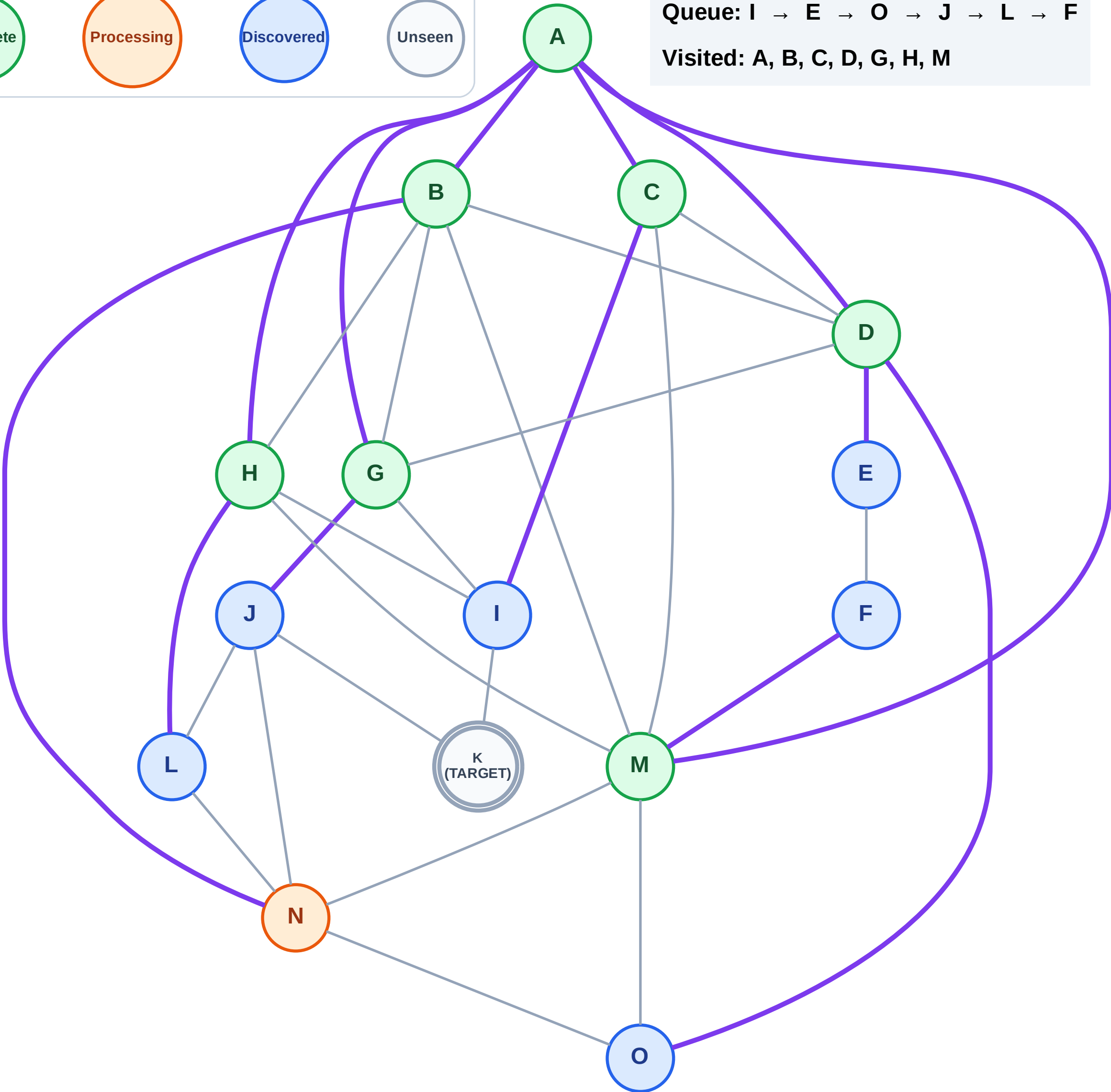
Processing

Discovered

Unseen

Queue: I → E → O → J → L → F

Visited: A, B, C, D, G, H, M



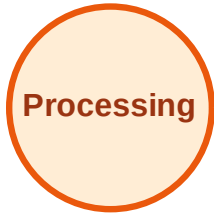
BFS Traversal

Step 17: No new neighbors from N

Legend



Complete



Processing



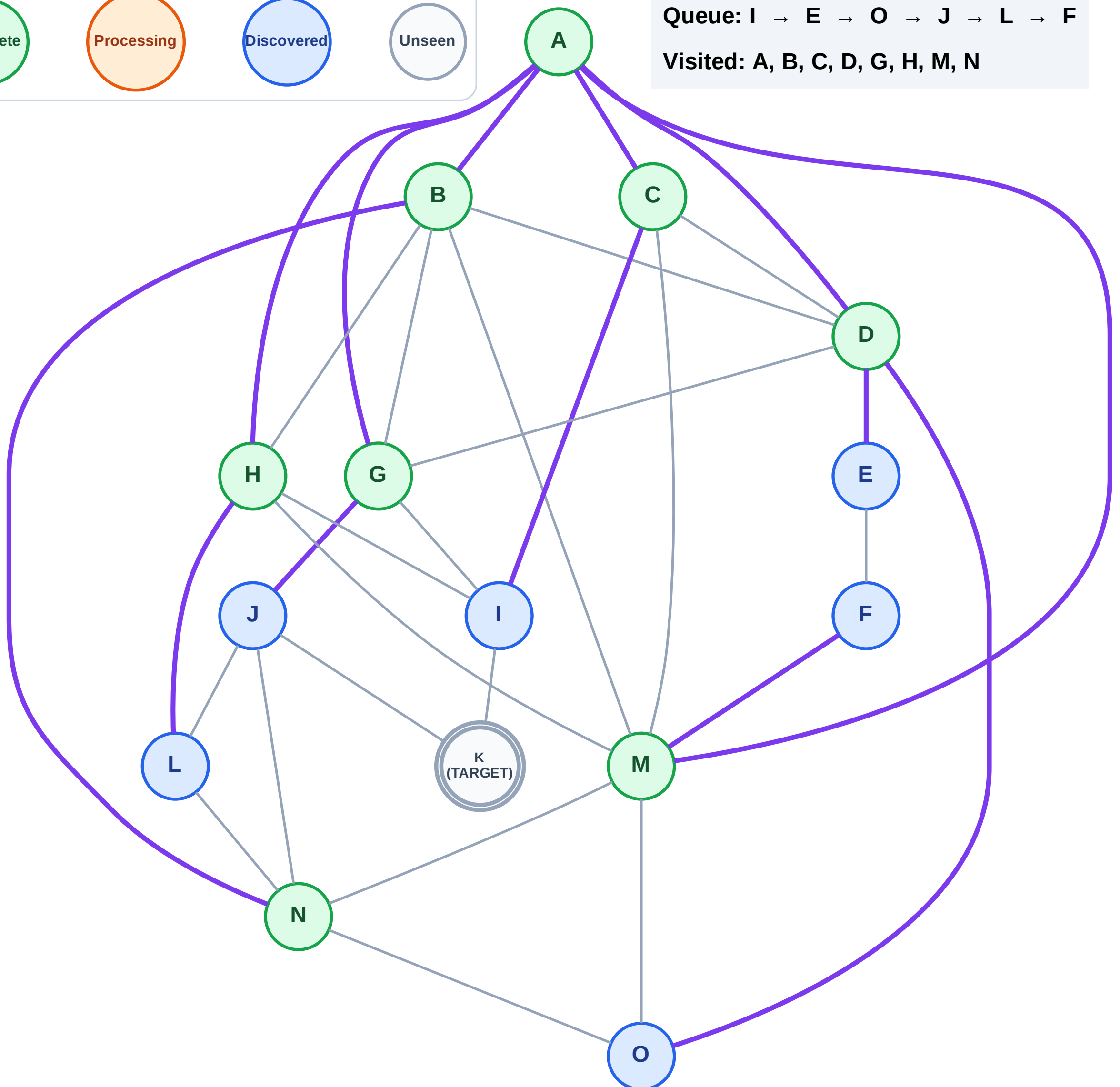
Discovered



Unseen

Queue: I → E → O → J → L → F

Visited: A, B, C, D, G, H, M, N



BFS Traversal

Step 18: Process node I

Legend

Complete

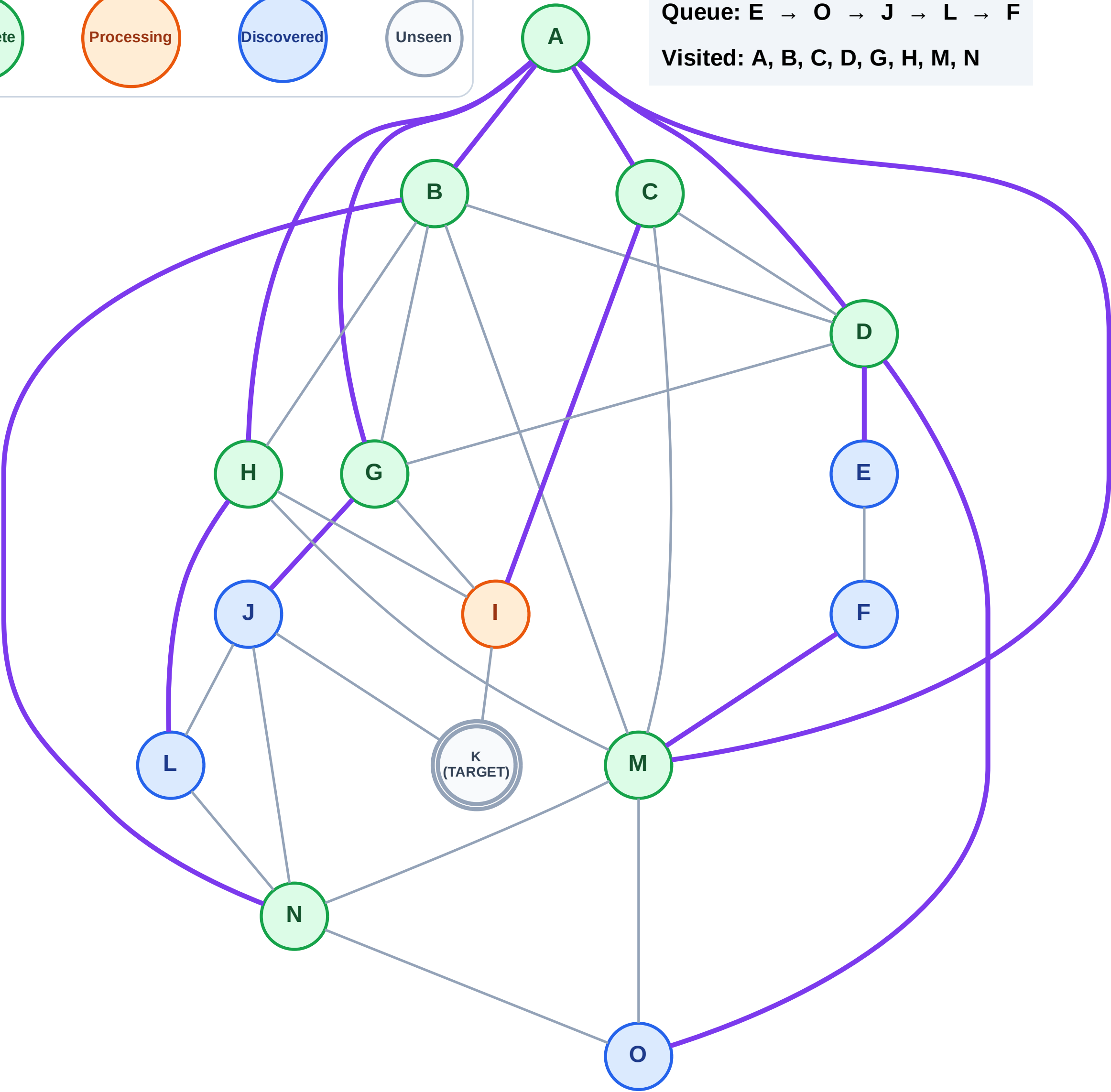
Processing

Discovered

Unseen

Queue: E → O → J → L → F

Visited: A, B, C, D, G, H, M, N



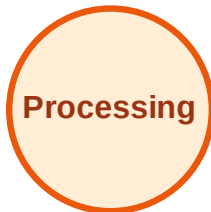
BFS Traversal

Step 19: Discover K from I

Legend



Complete



Processing



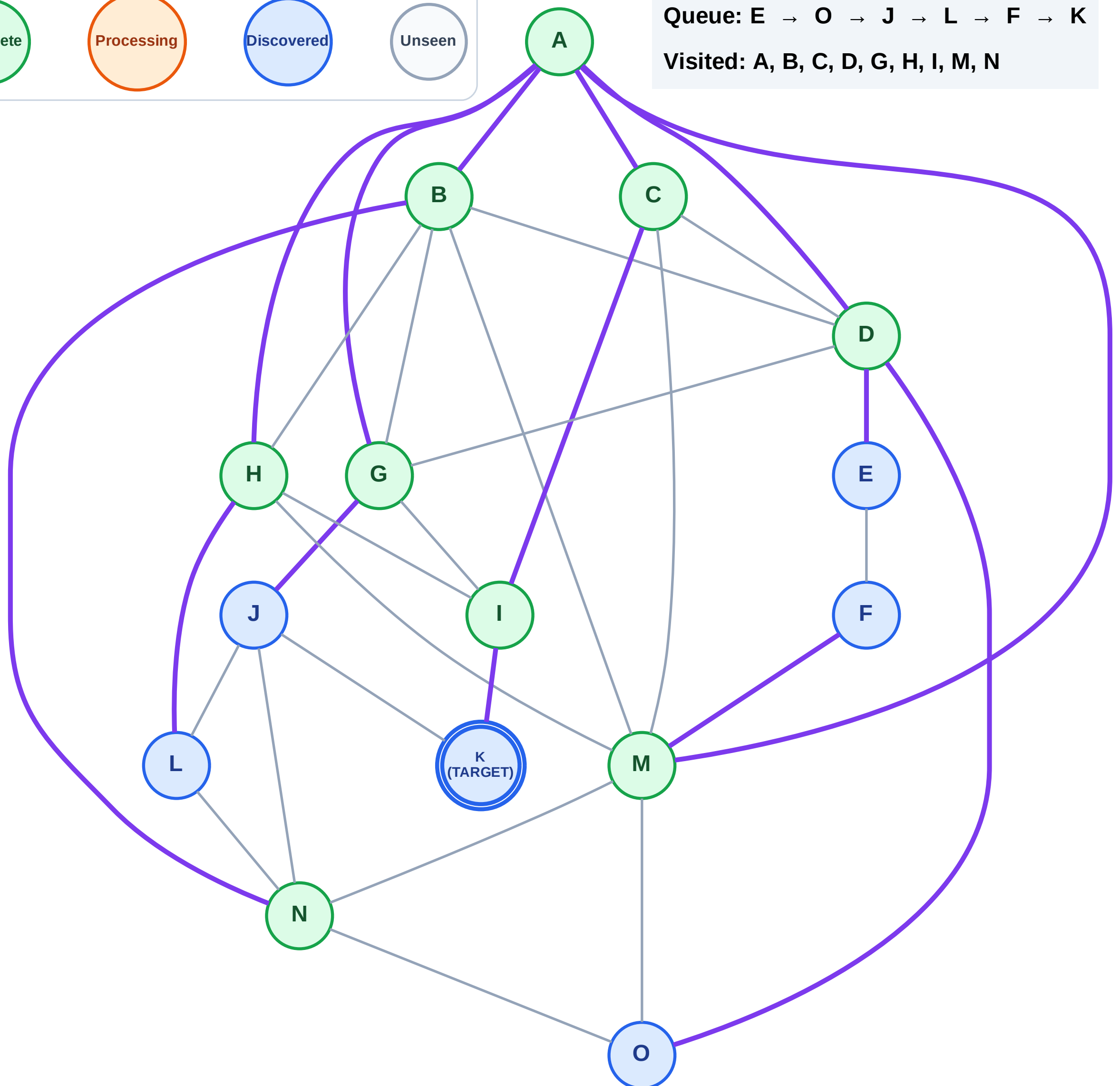
Discovered



Unseen

Queue: E → O → J → L → F → K

Visited: A, B, C, D, G, H, I, M, N



BFS Traversal

Step 20: Process node E

Legend

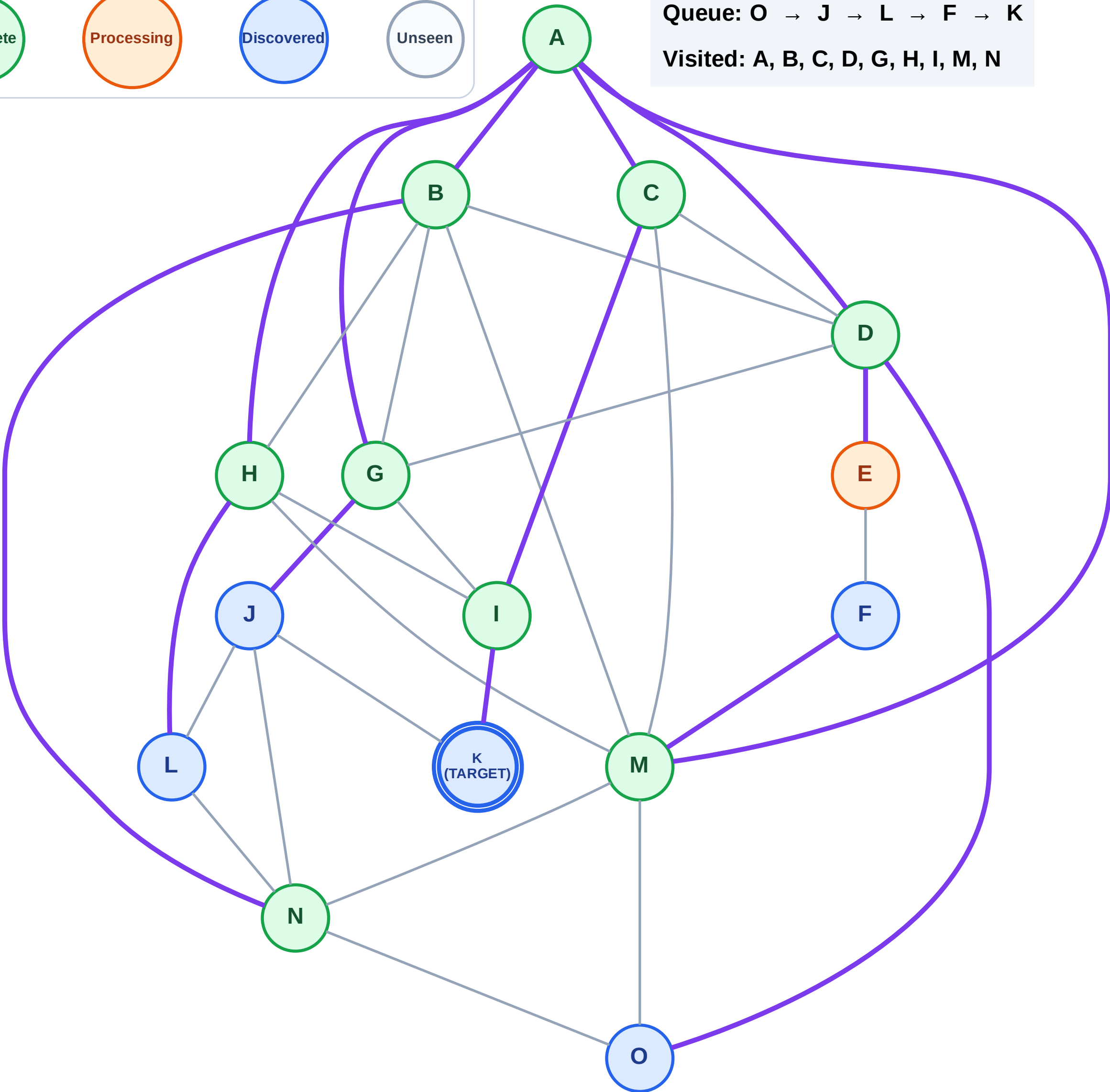
Complete

Processing

Discovered

Unseen

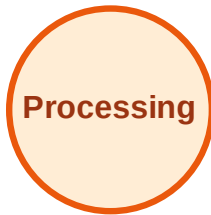
Queue: O → J → L → F → K
Visited: A, B, C, D, G, H, I, M, N



BFS Traversal

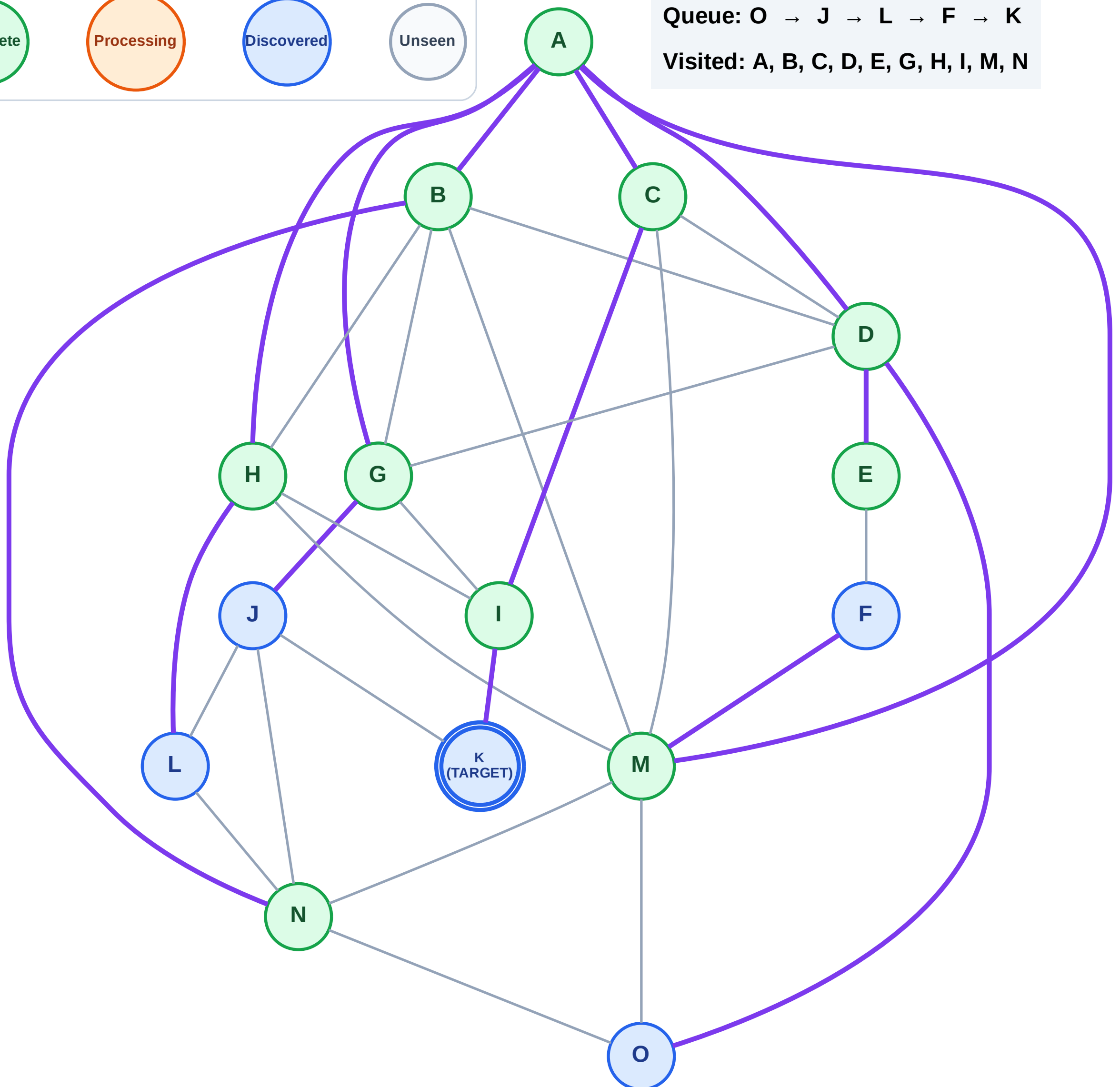
Step 21: No new neighbors from E

Legend



Queue: O → J → L → F → K

Visited: A, B, C, D, E, G, H, I, M, N



BFS Traversal

Step 22: Process node O

Legend

Complete

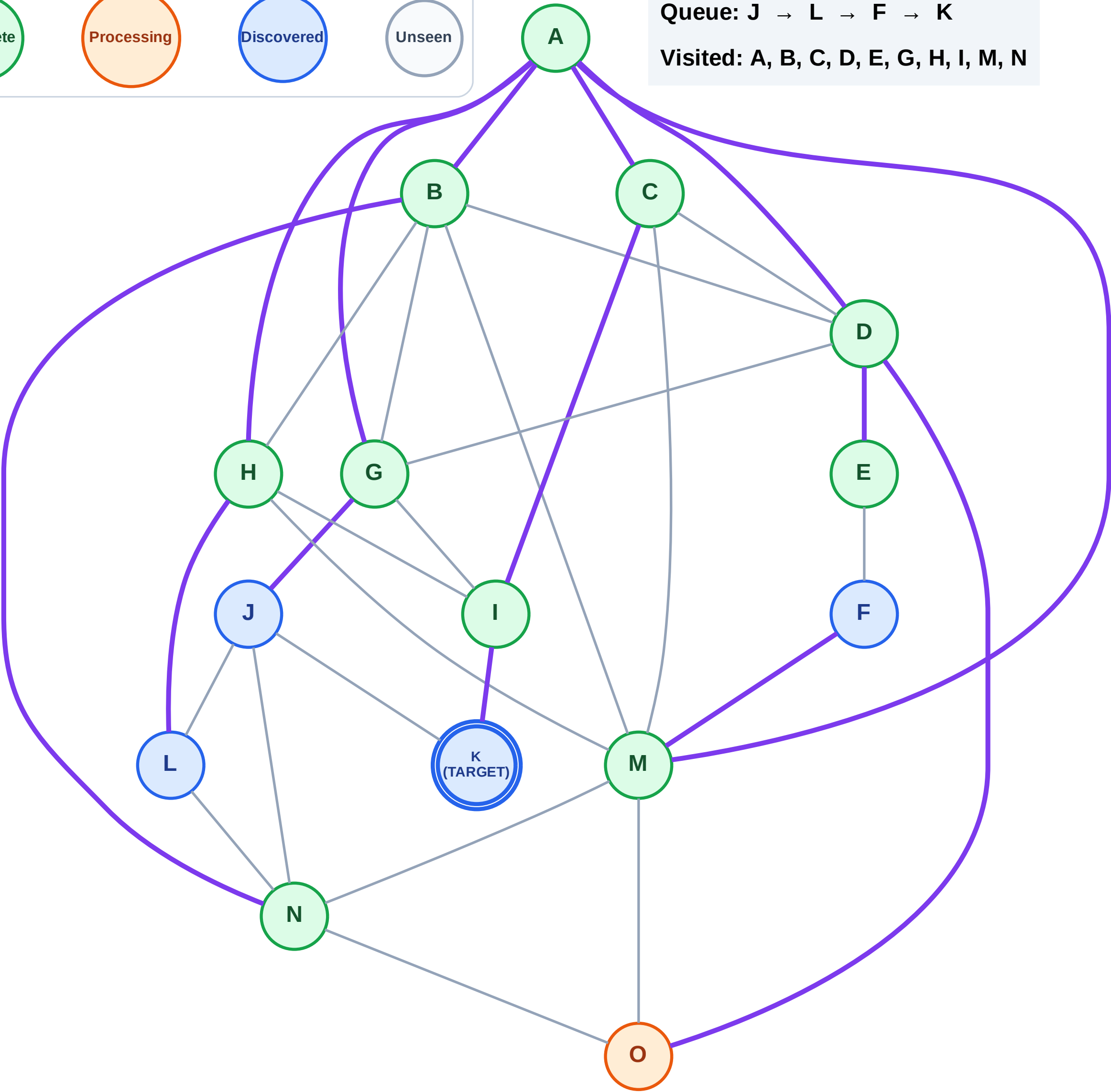
Processing

Discovered

Unseen

Queue: J → L → F → K

Visited: A, B, C, D, E, G, H, I, M, N



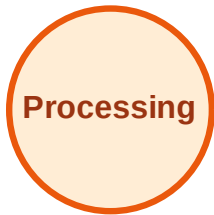
BFS Traversal

Step 23: No new neighbors from O

Legend



Complete



Processing



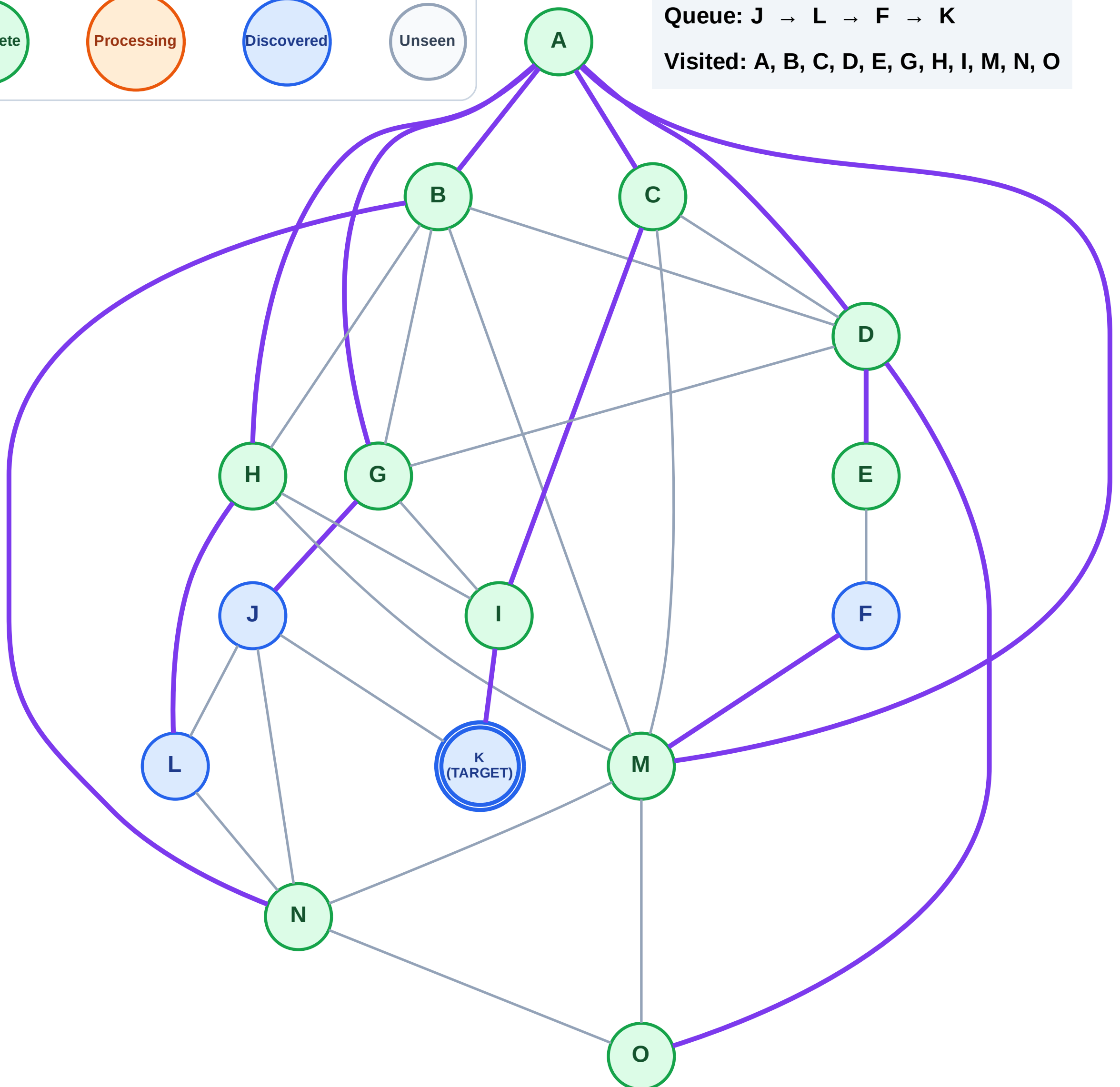
Discovered



Unseen

Queue: J → L → F → K

Visited: A, B, C, D, E, G, H, I, M, N, O



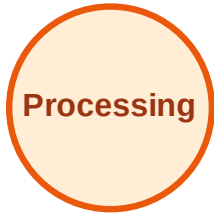
BFS Traversal

Step 24: Process node J

Legend



Complete



Processing



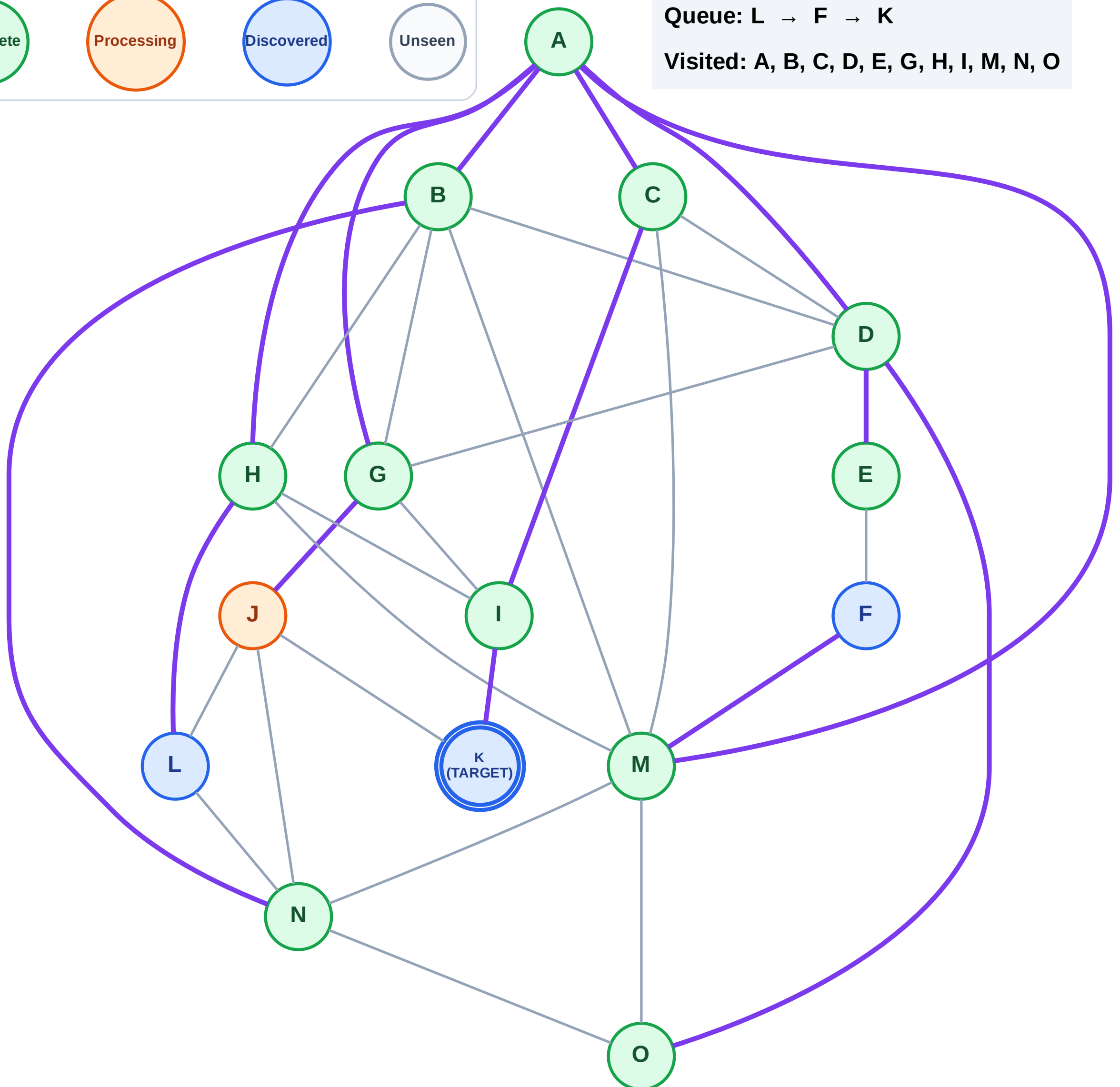
Discovered



Unseen

Queue: L → F → K

Visited: A, B, C, D, E, G, H, I, M, N, O



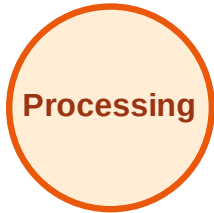
BFS Traversal

Step 25: No new neighbors from J

Legend



Complete



Processing



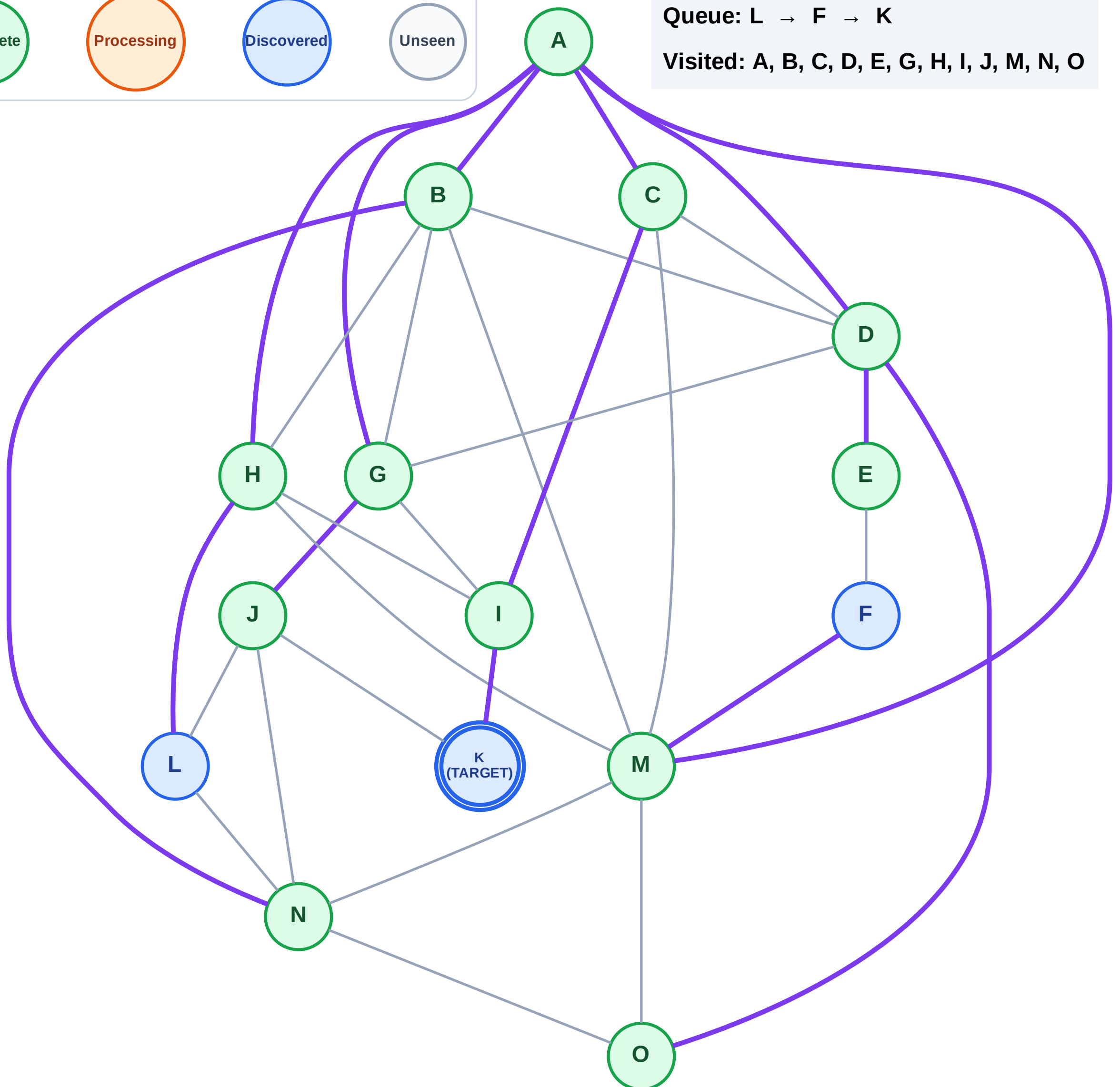
Discovered



Unseen

Queue: L → F → K

Visited: A, B, C, D, E, G, H, I, J, M, N, O



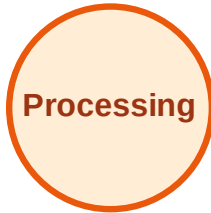
BFS Traversal

Step 26: Process node L

Legend



Complete



Processing



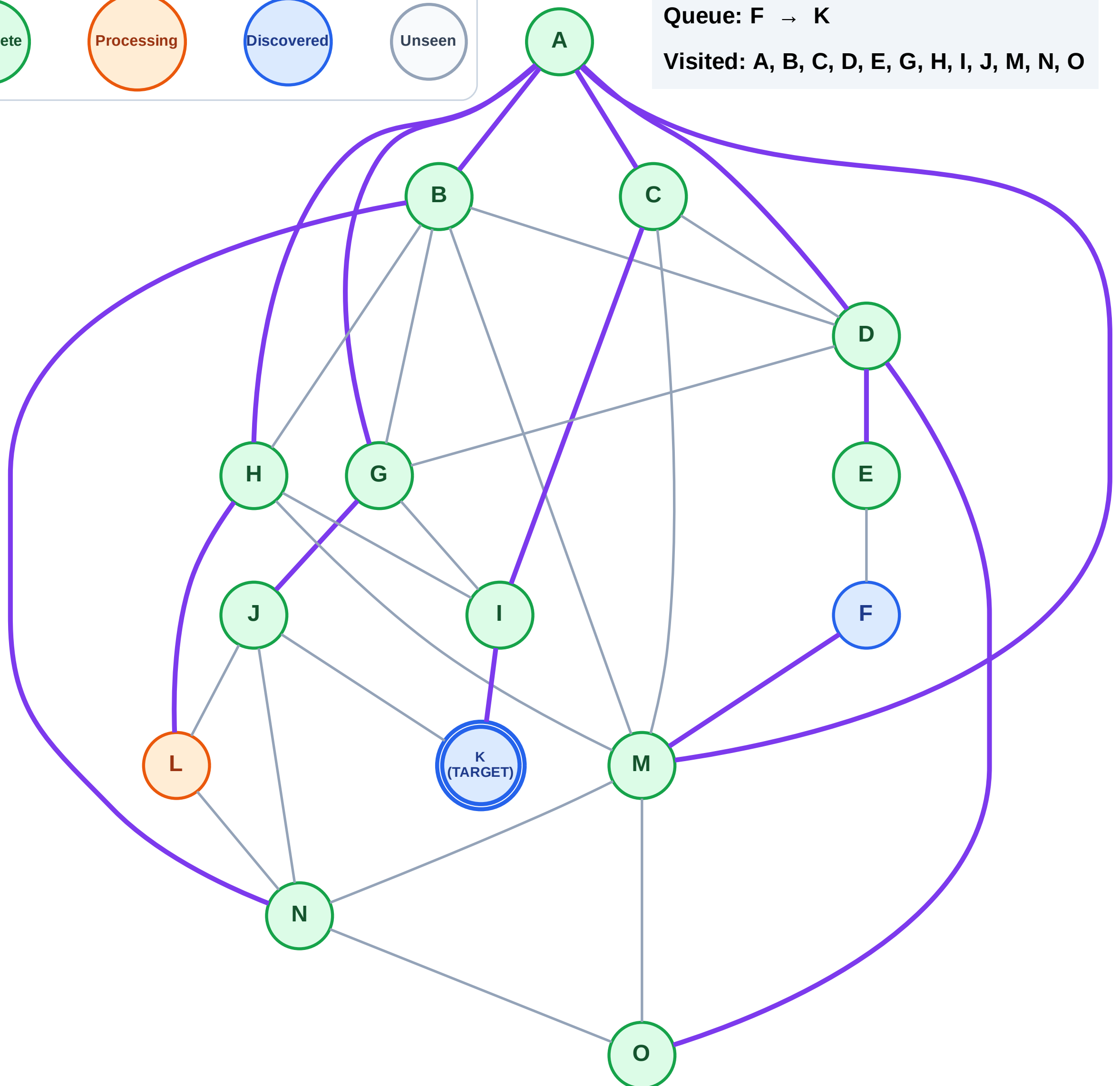
Discovered



Unseen

Queue: F → K

Visited: A, B, C, D, E, G, H, I, J, M, N, O



Step 27: No new neighbors from L

Step 27: No new neighbors from L



BFS Traversal

Step 28: Process node F

Legend

Complete

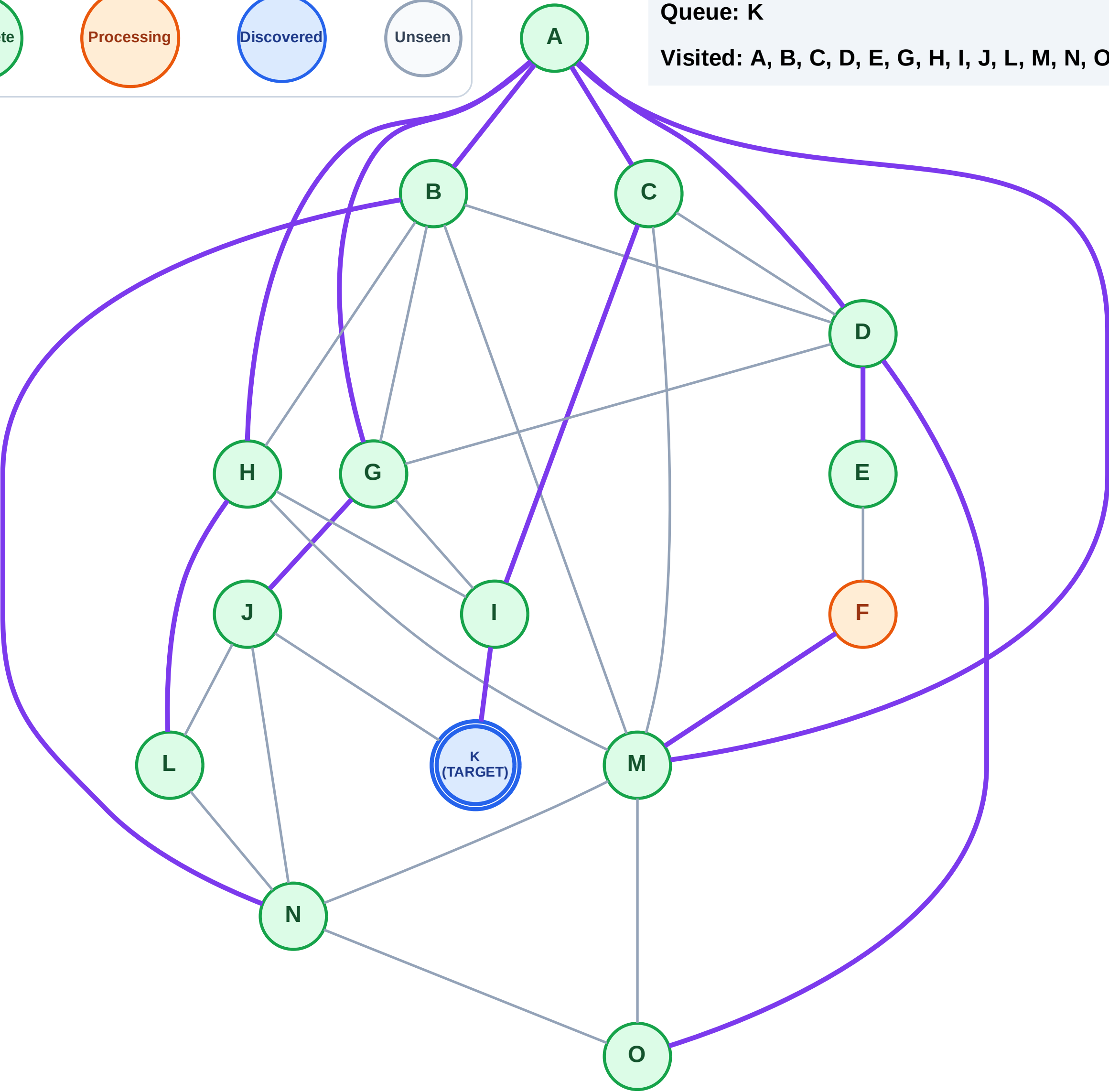
Processing

Discovered

Unseen

Queue: K

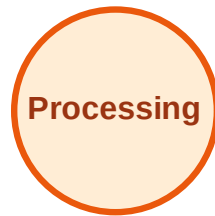
Visited: A, B, C, D, E, G, H, I, J, L, M, N, O



BFS Traversal

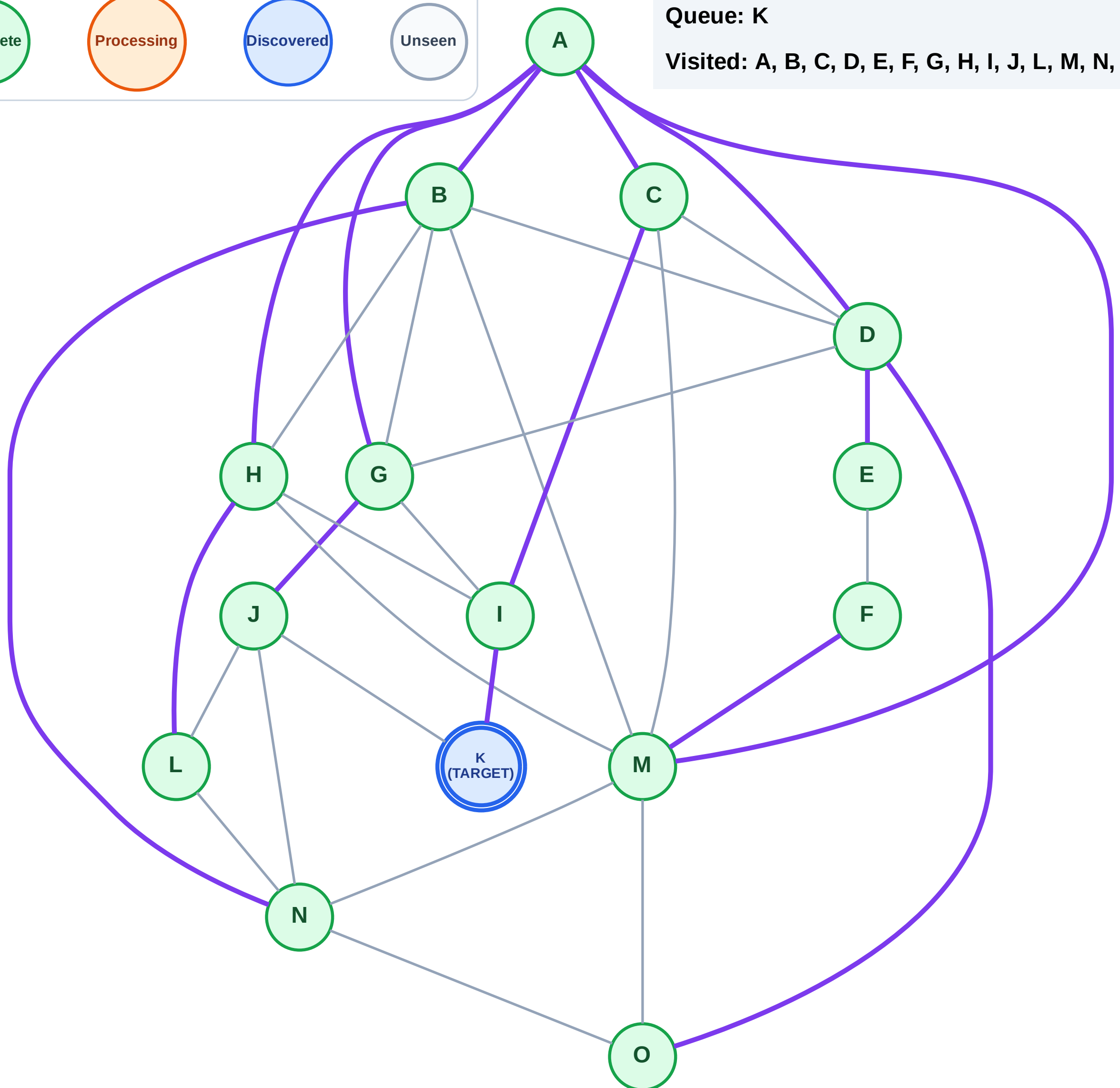
Step 29: No new neighbors from F

Legend

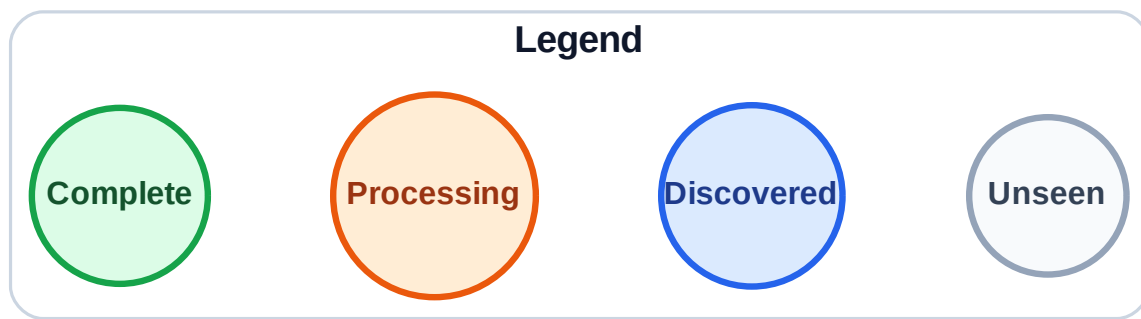


Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O

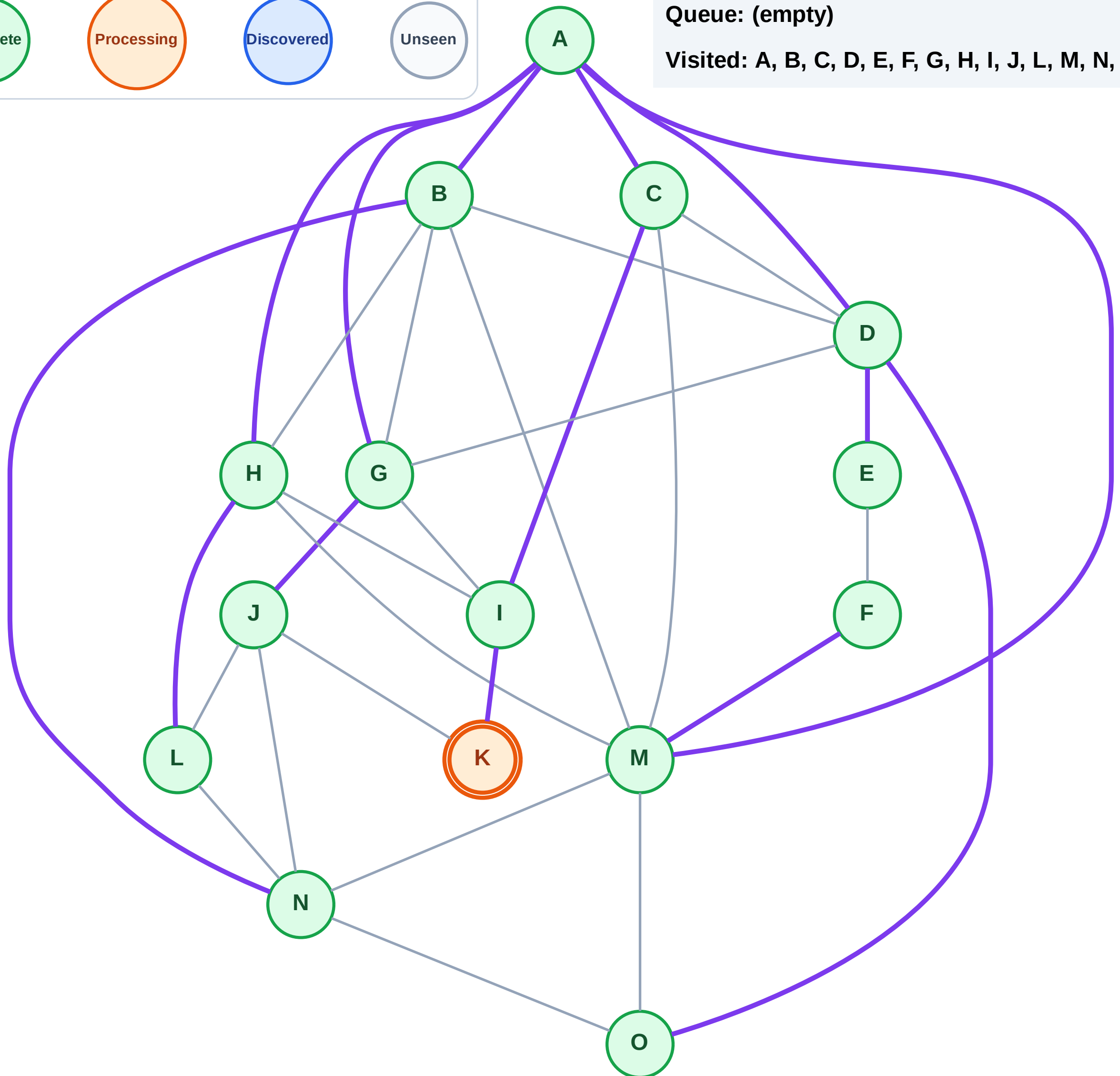


Step 30: Process node K



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 31: TARGET NODE K REACHED! Path (3 hops): A → C → I → K

Legend

Complete

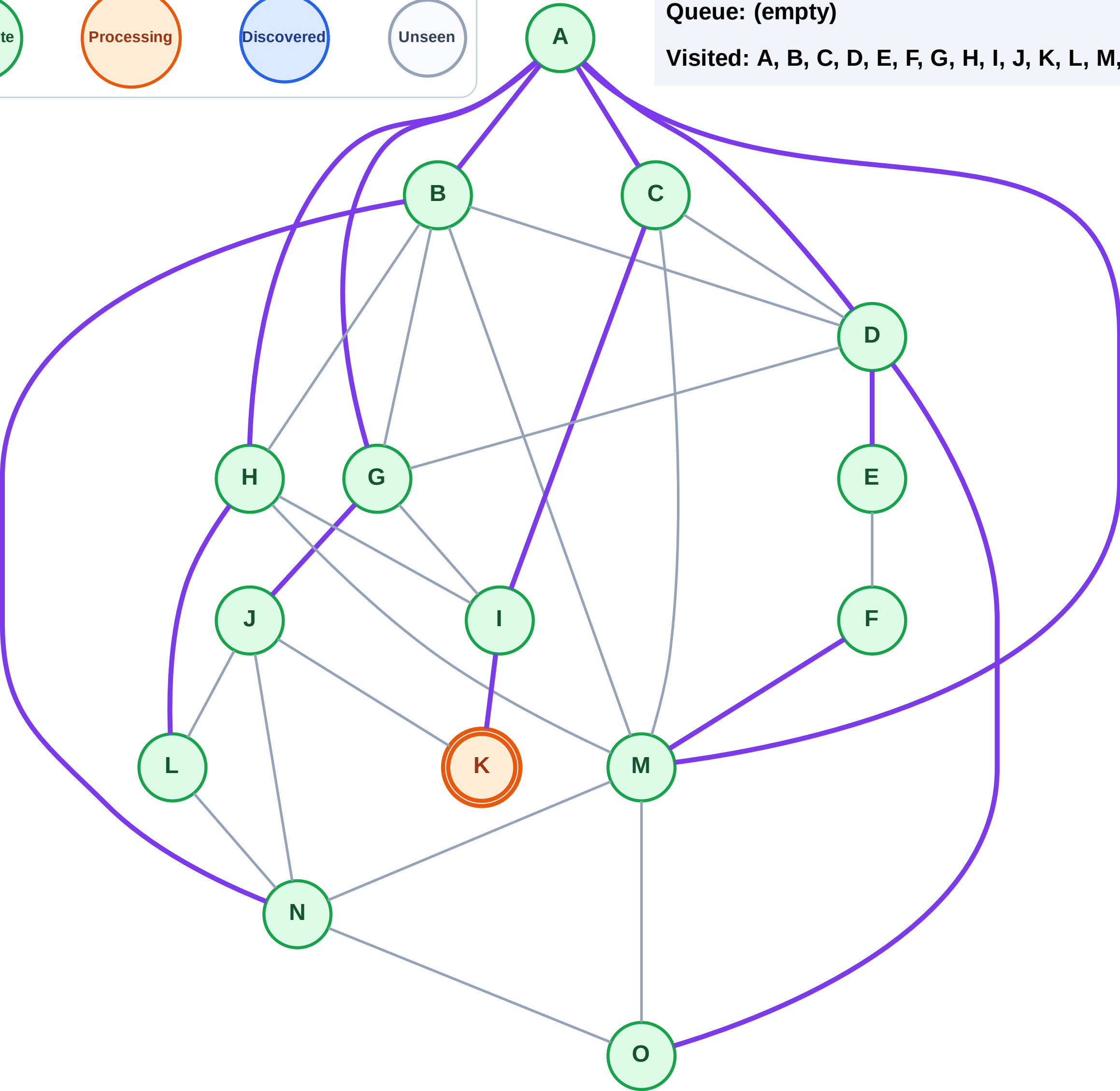
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



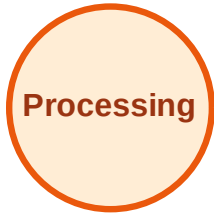
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



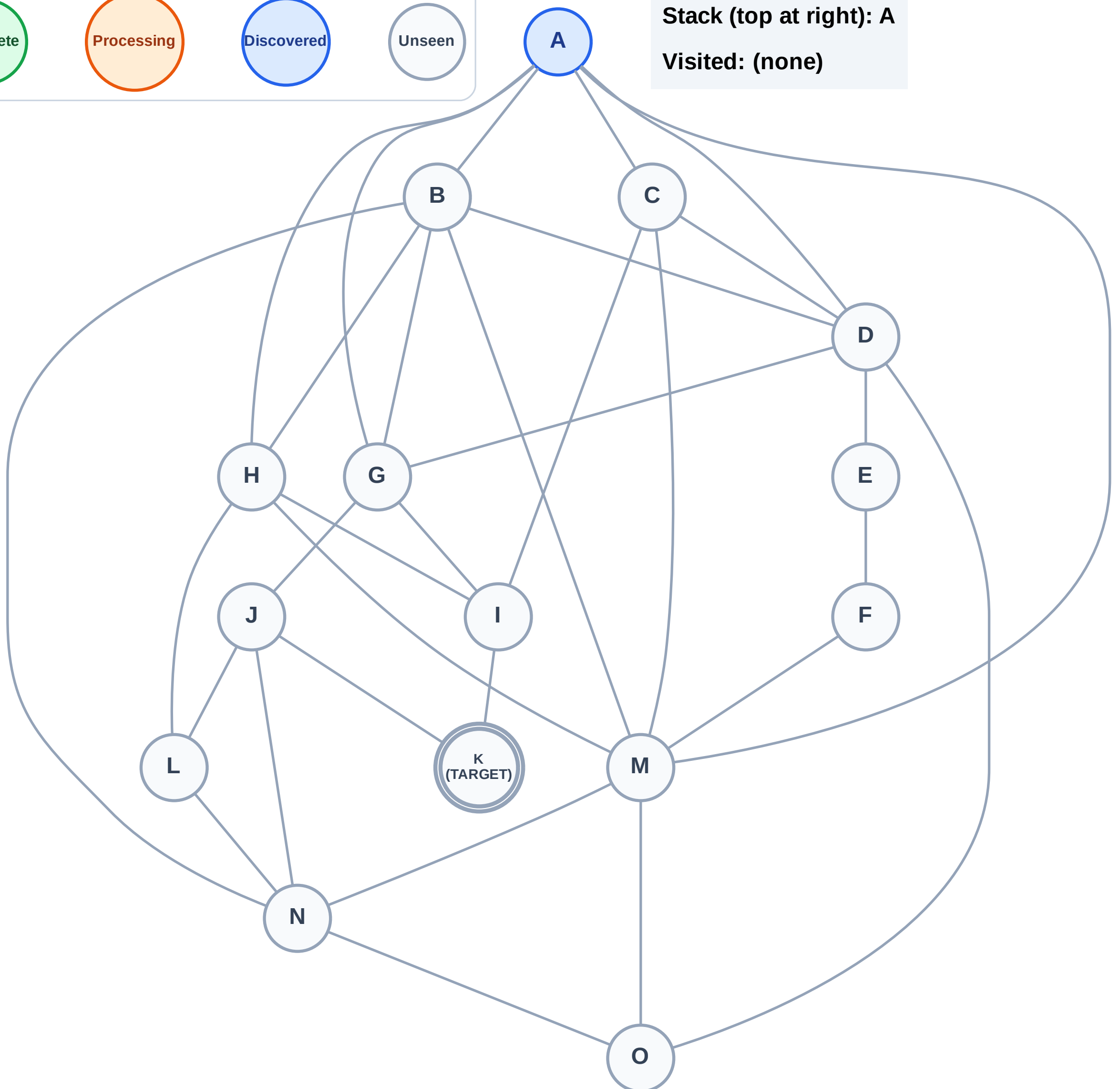
Discovered



Unseen

Stack (top at right): A

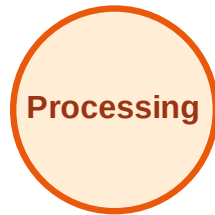
Visited: (none)



DFS Traversal

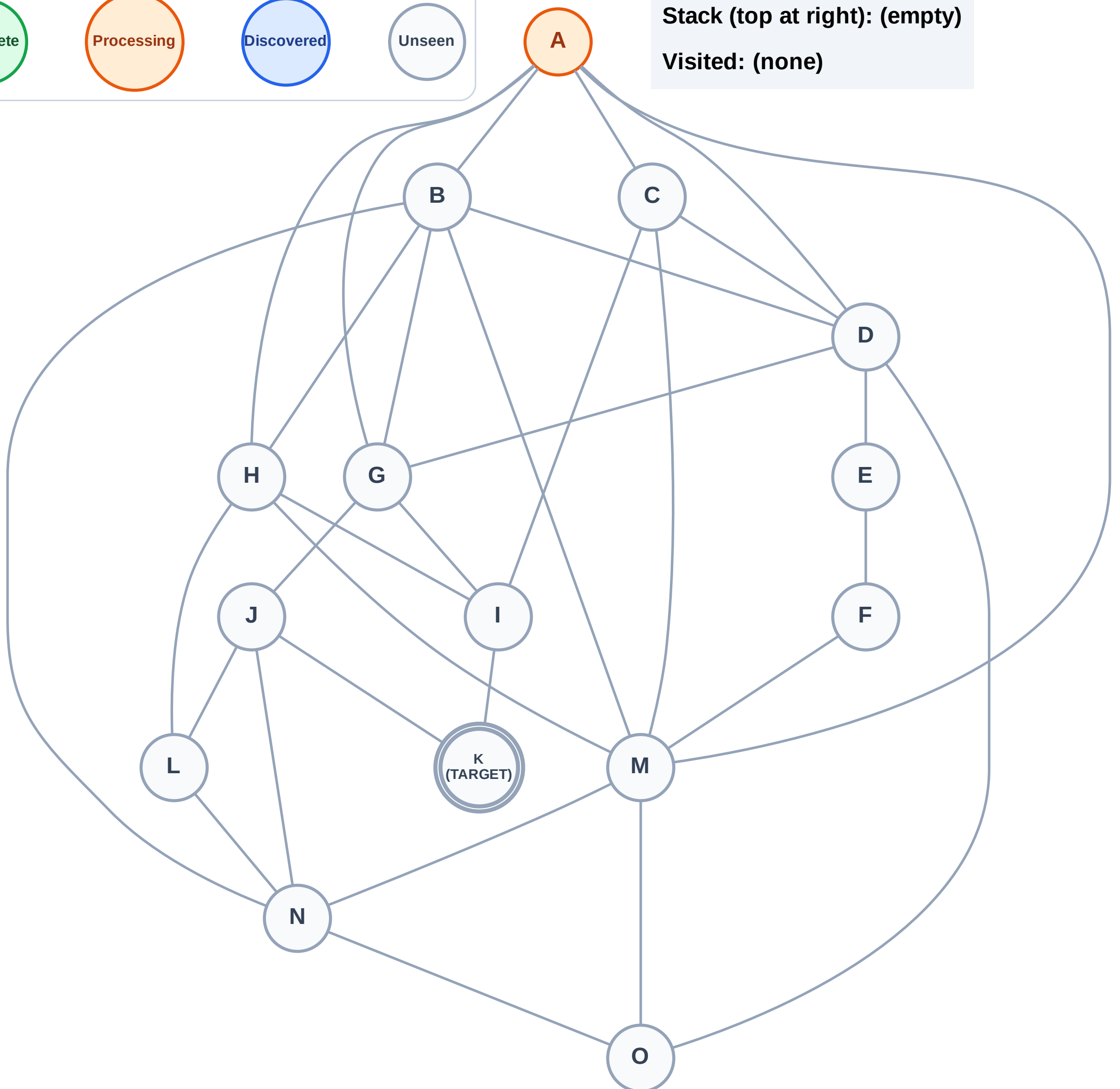
Step 2: Process node A

Legend



Stack (top at right): (empty)

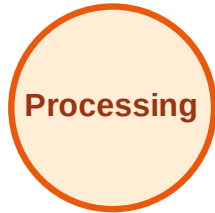
Visited: (none)



DFS Traversal

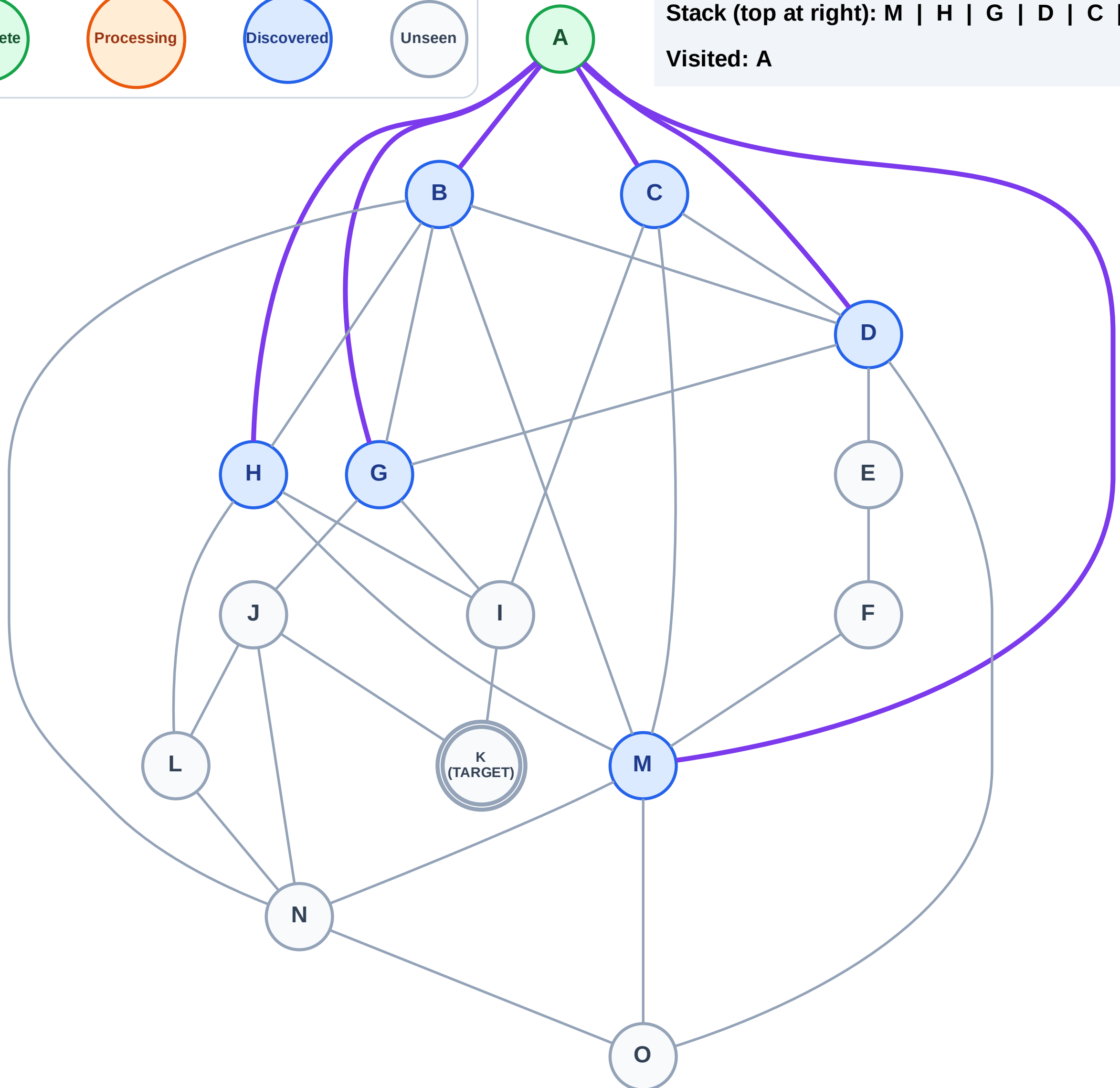
Step 3: Discover M, H, G, D, C, B from A

Legend



Stack (top at right): M | H | G | D | C | B

Visited: A



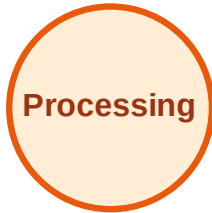
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



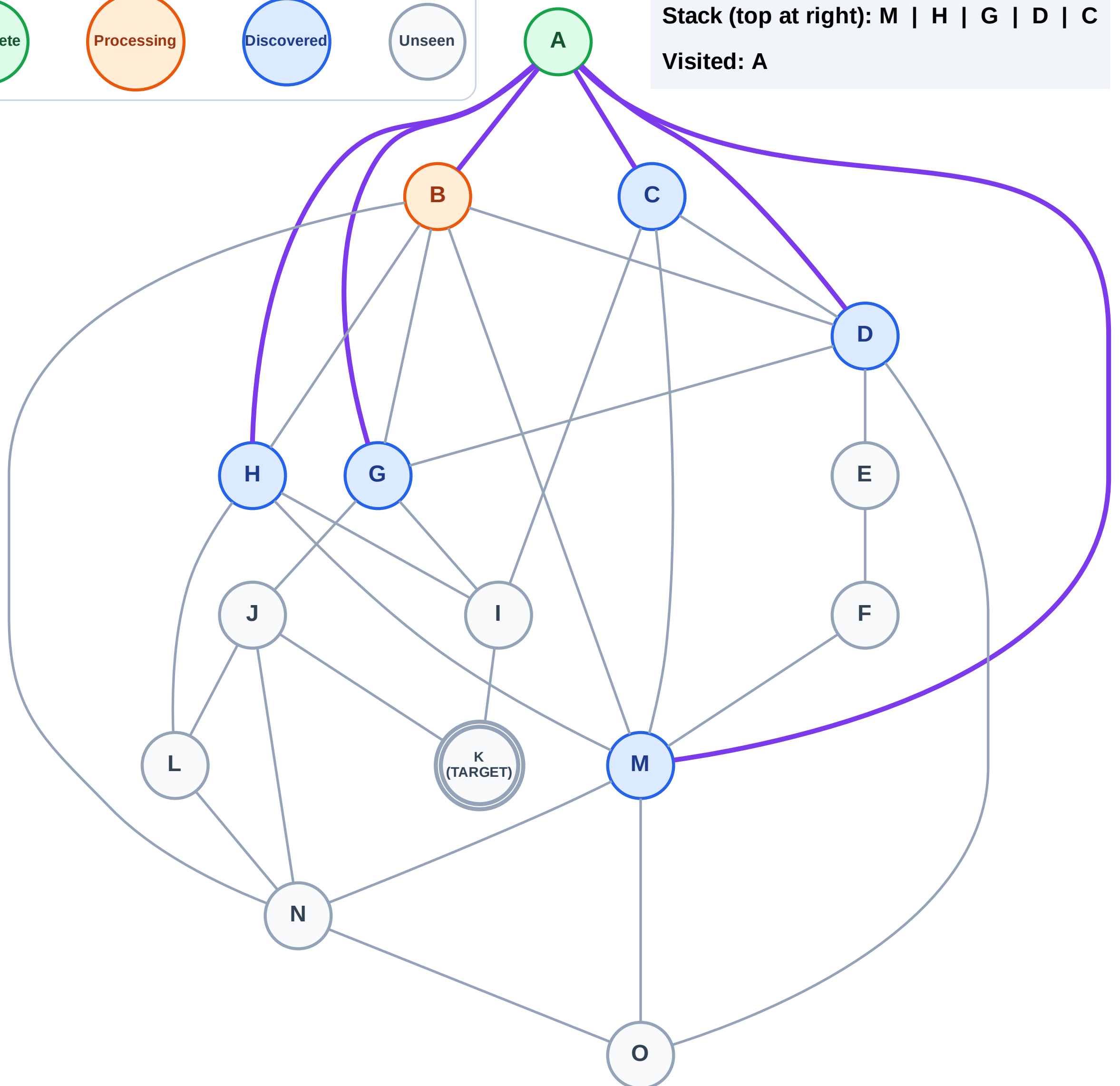
Discovered



Unseen

Stack (top at right): M | H | G | D | C

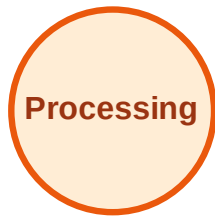
Visited: A



DFS Traversal

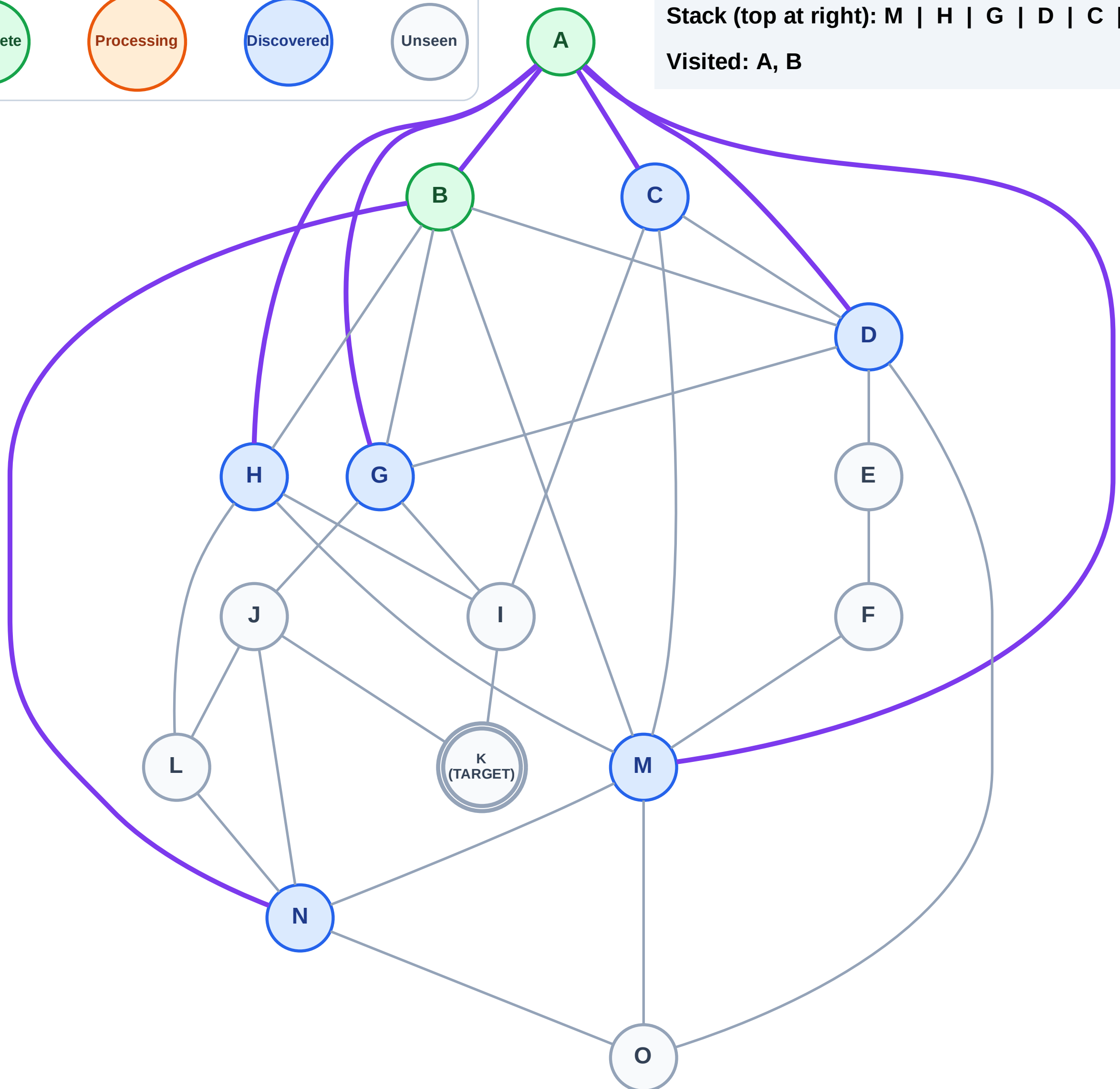
Step 5: Discover N from B

Legend



Stack (top at right): M | H | G | D | C | N

Visited: A, B



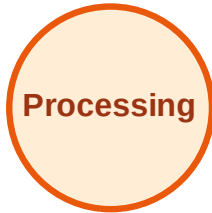
DFS Traversal

Step 6: Process node N

Legend



Complete



Processing



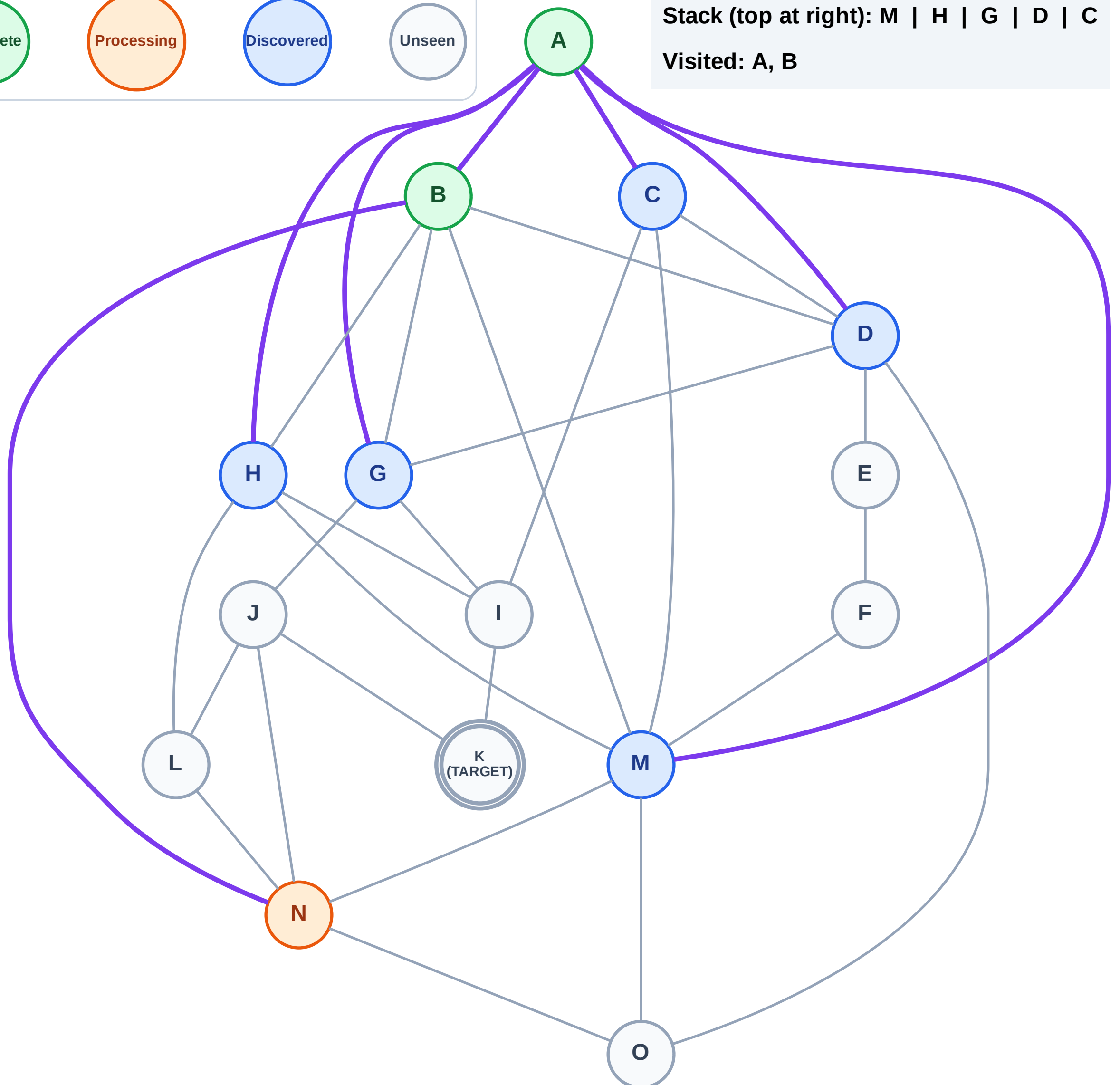
Discovered



Unseen

Stack (top at right): M | H | G | D | C

Visited: A, B



DFS Traversal

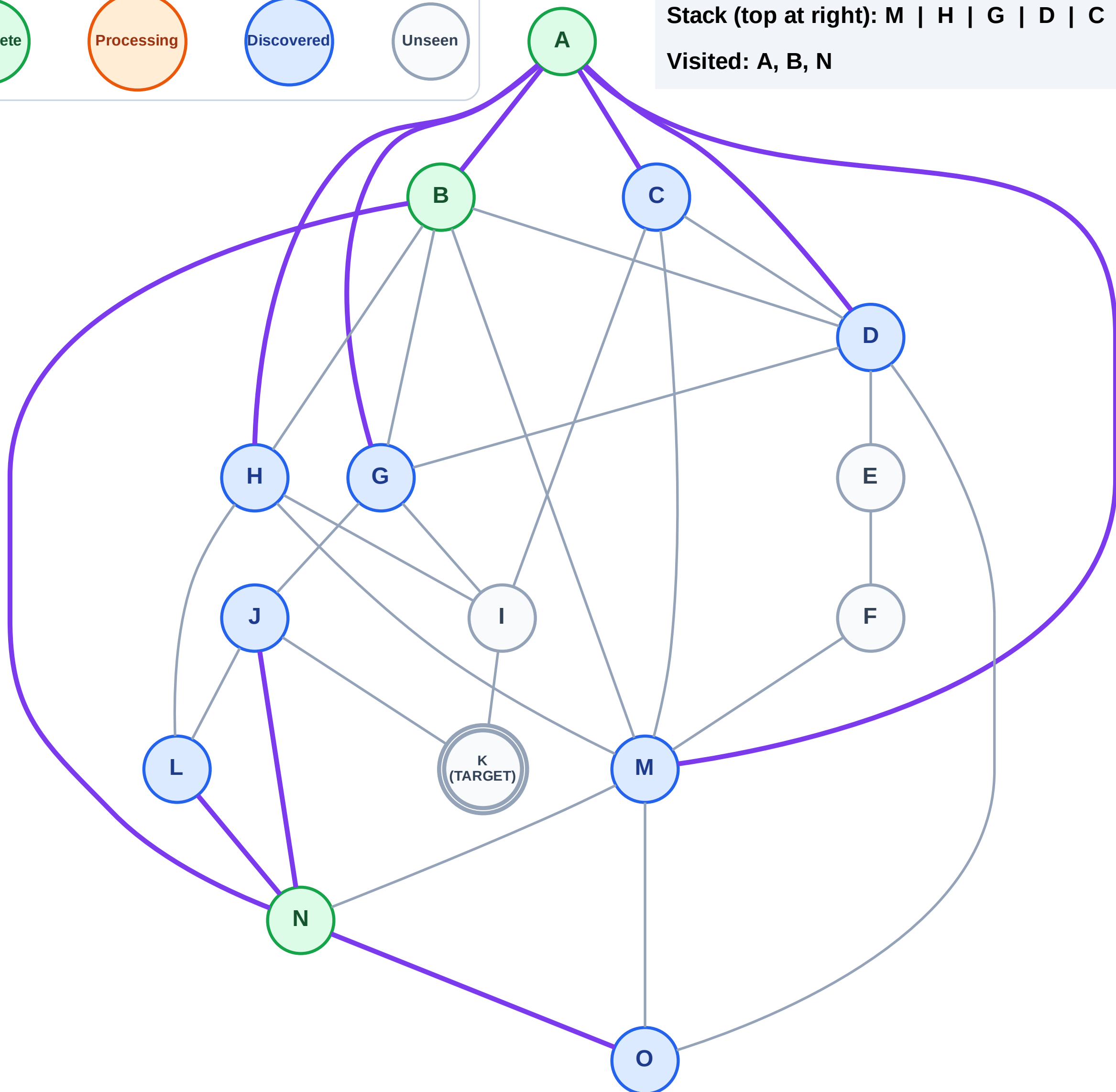
Step 7: Discover O, L, J from N

Legend



Stack (top at right): M | H | G | D | C | O | L | J

Visited: A, B, N



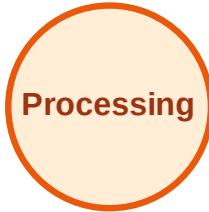
DFS Traversal

Step 8: Process node J

Legend



Complete



Processing



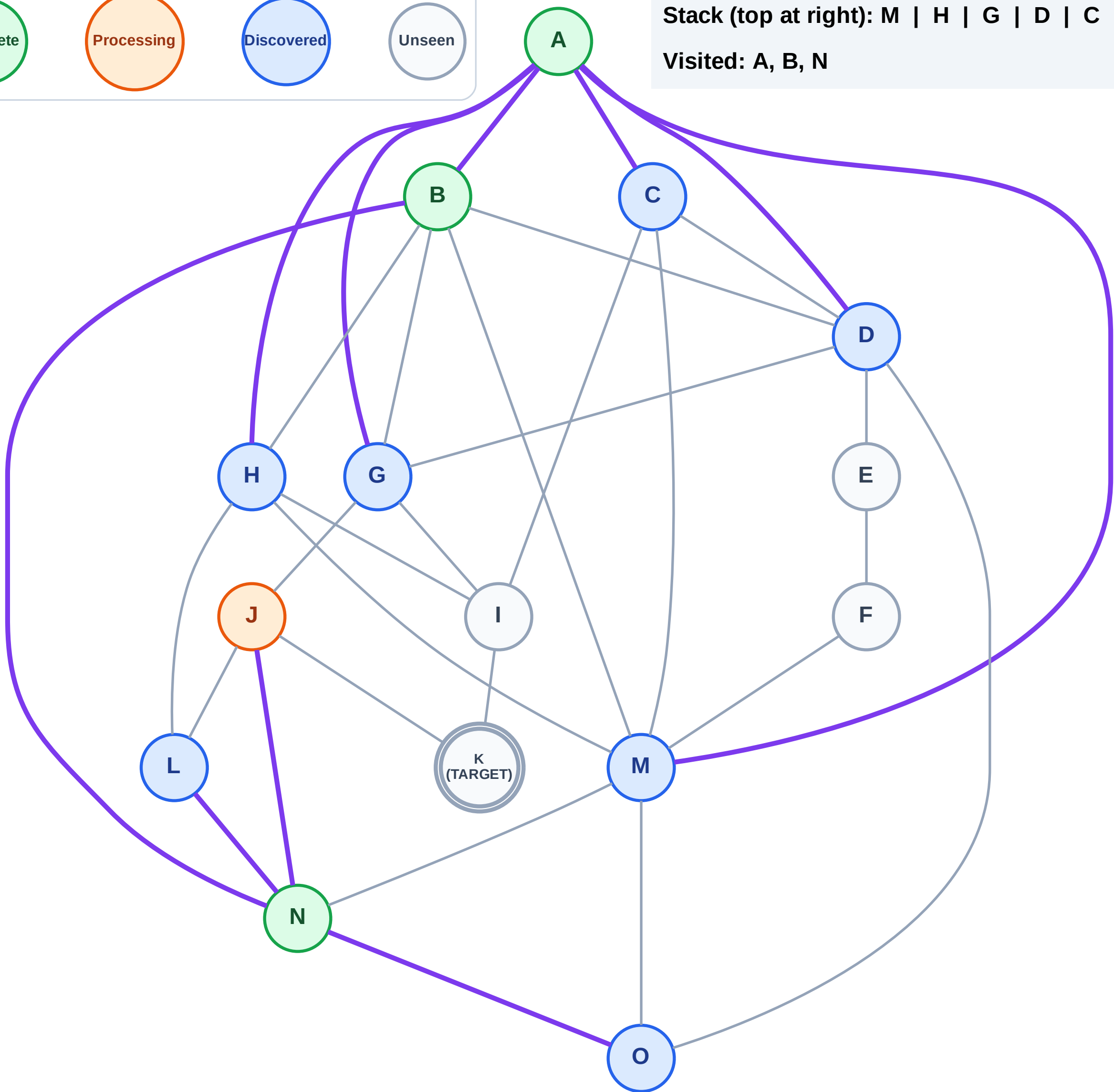
Discovered



Unseen

Stack (top at right): M | H | G | D | C | O | L

Visited: A, B, N



DFS Traversal

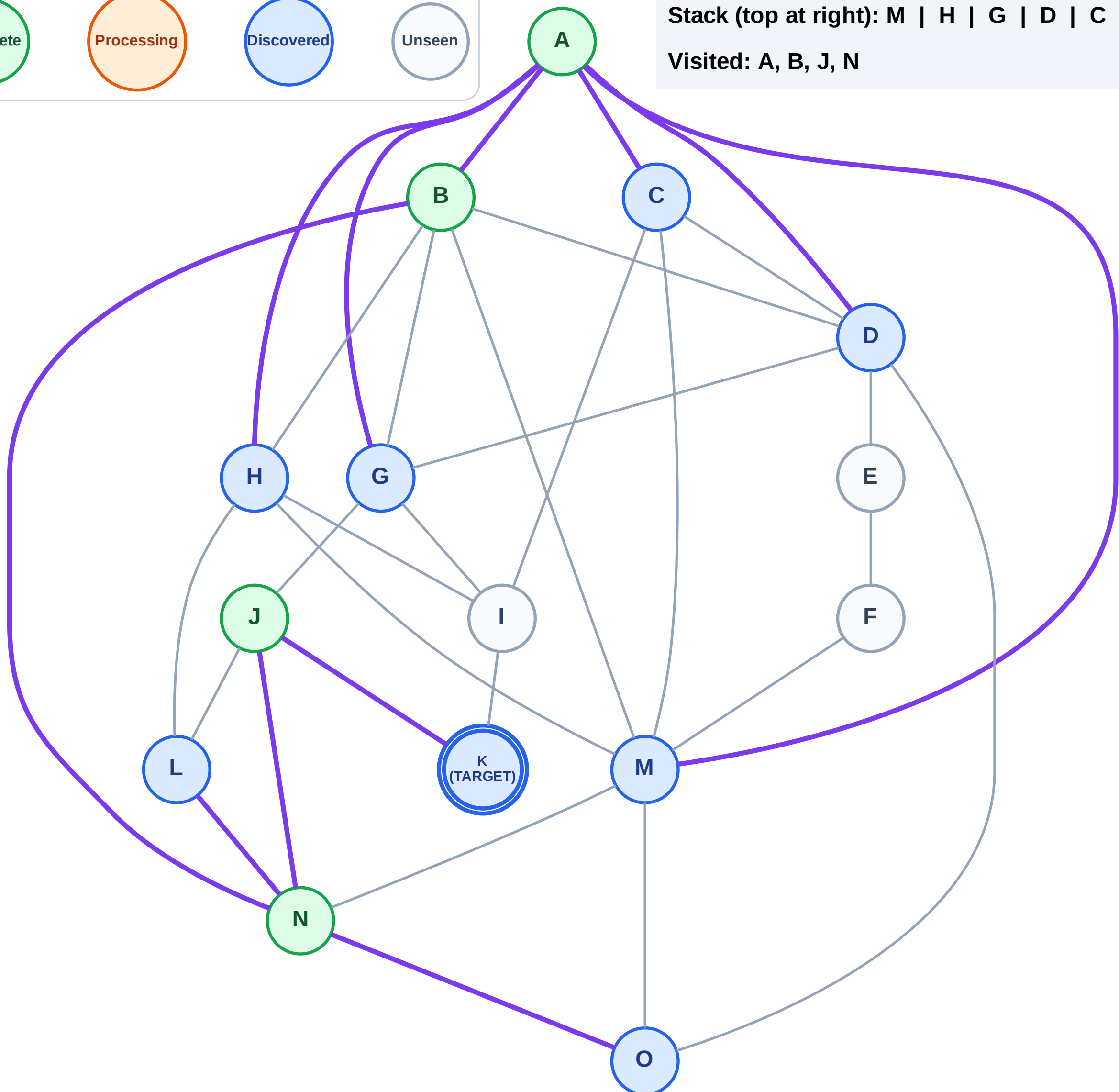
Step 9: Discover K from J

Legend



Stack (top at right): M | H | G | D | C | O | L | K

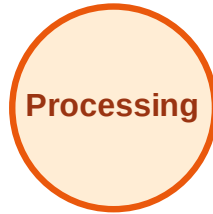
Visited: A, B, J, N



DFS Traversal

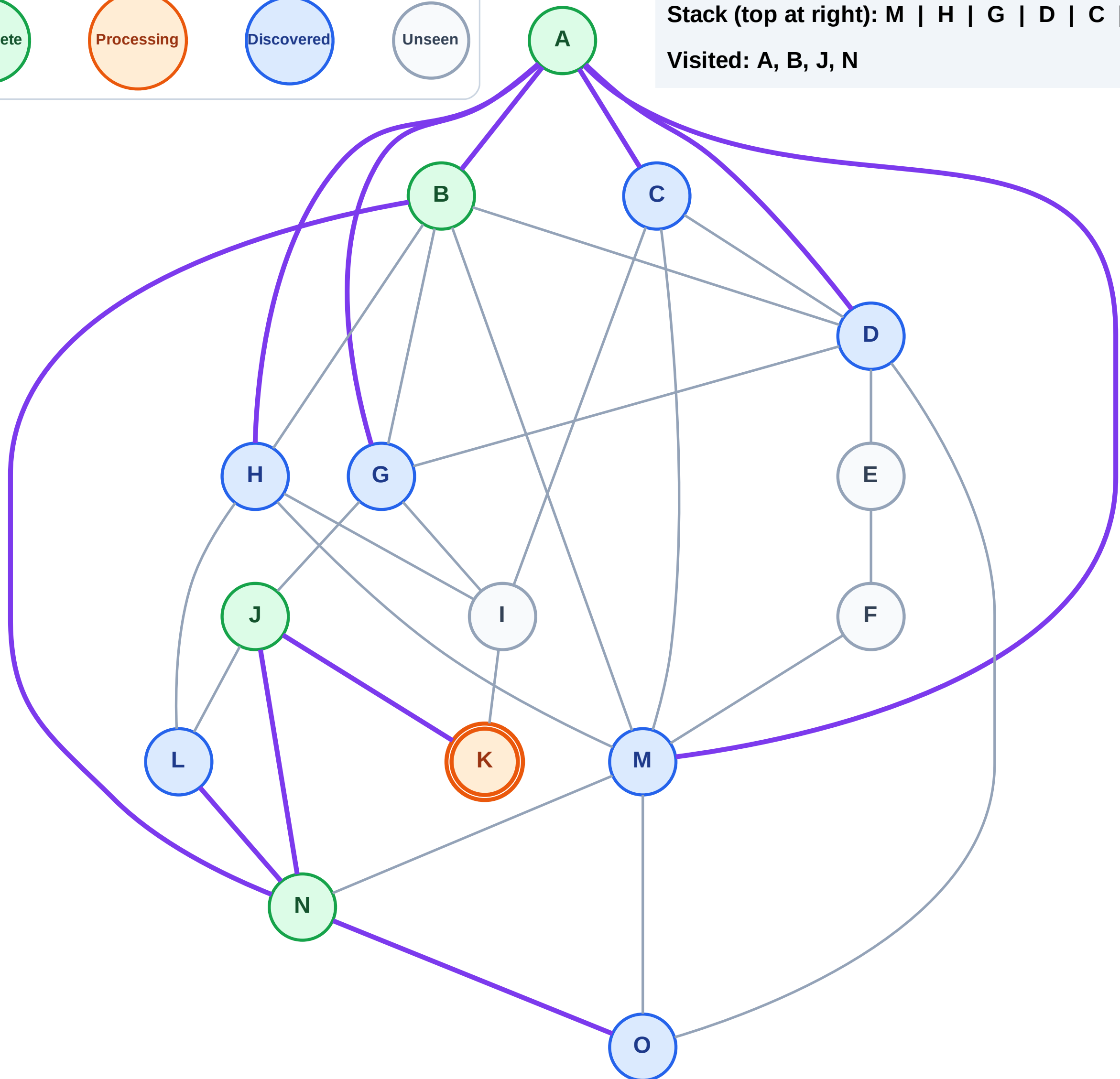
Step 10: Process node K

Legend



Stack (top at right): M | H | G | D | C | O | L

Visited: A, B, J, N



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (4 hops): A → B → N → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | H | G | D | C | O | L

Visited: A, B, J, K, N

